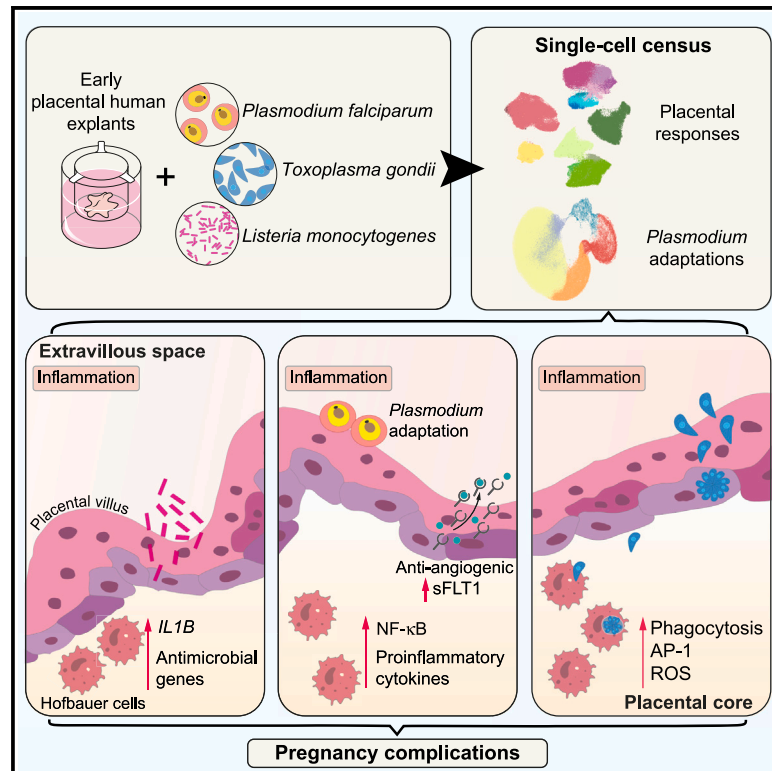


Acute response to pathogens in the early human placenta at single-cell resolution

Graphical abstract



Authors

Regina Hoo, Elias R. Ruiz-Morales, Iva Kelava, ..., Damiana Álvarez-Errico, Marcus C.S. Lee, Roser Vento-Tormo

Correspondence

mlee001@dundee.ac.uk (M.C.S.L.), rv4@sanger.ac.uk (R.V.-T.)

In brief

Placental infections during early pregnancy can lead to pregnancy complications and developmental anomalies. Hoo, Ruiz-Morales, Kelava et al. profiled the placenta at the single-cell level upon exposure to *Listeria*, *Plasmodium*, and *Toxoplasma*. They uncovered general inflammation and unique Hofbauer cell responses. Additionally, they revealed *Plasmodium* adaptation to the placental microenvironment.

Highlights

- Single-cell census of early placental human explants exposed to congenital pathogens
- Infections trigger a general inflammatory response across all placental lineages
- Hofbauer cells elicit pathogen-specific responses and demonstrate antimicrobial abilities
- *Plasmodium* adapts to the placenta through erythrocyte remodeling and nutrient uptake



Article

Acute response to pathogens in the early human placenta at single-cell resolution

Regina Hoo,^{1,2,8} Elias R. Ruiz-Morales,^{1,8} Iva Kelava,^{1,8} Mukul Rawat,^{1,3} Cecilia Icoresi Mazzeo,¹ Elizabeth Tuck,¹ Carmen Sancho-Serra,¹ Sara Chelaghma,⁴ Alexander V. Predeus,¹ Simon Murray,¹ David Fernandez-Antoran,^{5,6} Ross F. Waller,⁴ Damiana Álvarez-Errico,⁷ Marcus C.S. Lee,^{1,3,*} and Roser Vento-Tormo^{1,2,9,*}

¹Wellcome Sanger Institute, Cambridge, UK

²Centre for Trophoblast Research, University of Cambridge, Cambridge, UK

³Wellcome Centre for Anti-Infectives Research, Division of Biological Chemistry and Drug Discovery, University of Dundee, Dundee, UK

⁴Department of Biochemistry, University of Cambridge, Cambridge, UK

⁵Wellcome/Cancer Research UK Gurdon Institute, University of Cambridge, Cambridge, UK

⁶Department of Pathology, University of Cambridge, Cambridge, UK

⁷IMIDomics, Inc., Barcelona, Spain

⁸These authors contributed equally

⁹Lead contact

*Correspondence: mlee001@dundee.ac.uk (M.C.S.L.), rv4@sanger.ac.uk (R.V.-T.)

<https://doi.org/10.1016/j.cels.2024.04.002>

SUMMARY

The placenta is a selective maternal-fetal barrier that provides nourishment and protection from infections. However, certain pathogens can attach to and even cross the placenta, causing pregnancy complications with potential lifelong impacts on the child's health. Here, we profiled at the single-cell level the placental responses to three pathogens associated with intrauterine complications—*Plasmodium falciparum*, *Listeria monocytogenes*, and *Toxoplasma gondii*. We found that upon exposure to the pathogens, all placental lineages trigger inflammatory responses that may compromise placental function. Additionally, we characterized the responses of fetal macrophages known as Hofbauer cells (HBCs) to each pathogen and propose that they are the probable niche for *T. gondii*. Finally, we revealed how *P. falciparum* adapts to the placental microenvironment by modulating protein export into the host erythrocyte and nutrient uptake pathways. Altogether, we have defined the cellular networks and signaling pathways mediating acute placental inflammatory responses that could contribute to pregnancy complications.

INTRODUCTION

The placenta, the first organ generated by the embryo, serves as a selective barrier during pregnancy, allowing nutrition while preventing pathogens and toxic substances from crossing from mother to fetus. Despite this, certain pathogens, such as the eukaryotic parasites *Plasmodium falciparum* (*P. falciparum*) and *Toxoplasma gondii* (*T. gondii*) and the bacteria *Listeria monocytogenes* (*L. monocytogenes*), can cause pregnancy complications, leading to miscarriage, premature labor, low birth weight, and stillbirth.^{1–3} Pregnancy complications can be caused by the pathogens crossing or attaching to the placenta. For example, *P. falciparum*^{1,4} attaches to the placenta whereas others, such as *T. gondii*,⁵ *L. monocytogenes*,⁶ and some viruses,^{7–10} can cross it.^{1–3} When a pathogen such as *T. gondii* crosses the placenta, it can also lead to lifelong malformations in the brain of the offspring.³

The first-trimester embryo is particularly vulnerable to infections due to the adaptive immune response not being fully established.¹¹ Its protection relies on the innate responses in the

placenta, which effectively limit pathogen entry, as evidenced by the few types of pathogens that can cross it and their low transmission rates.¹² Trophoblasts, the epithelial-like cells of the placenta, form a two-layered tree-like structure that provides resistance against the majority of infections.^{9,13} In direct contact with the maternal blood is the syncytiotrophoblast (SCT), a multinucleated layer formed by the fusion of the villous cytotrophoblast (VCT) cells found beneath it. The SCT layer serves as a barrier, preventing the free passage of pathogens between maternal and fetal circulation.^{12,14,15}

In addition to trophoblasts, immune cells provide another layer of defense. Fetal macrophages, known as Hofbauer cells (HBCs),¹⁶ are the only immune cells present within the stroma of a healthy placenta.¹⁷ HBCs originate from the primitive hematopoietic wave¹⁸ and are typically considered tolerogenic and involved in angiogenesis.^{17,19} Earlier studies identified an antimicrobial role for HBCs in direct *in vitro* infection with *L. monocytogenes*,²⁰ but their role in more physiological systems remains unclear. In addition, while adult macrophages are known to play various roles in infection (e.g., phagocytizing pathogens^{21–24} and infected cells,²⁵



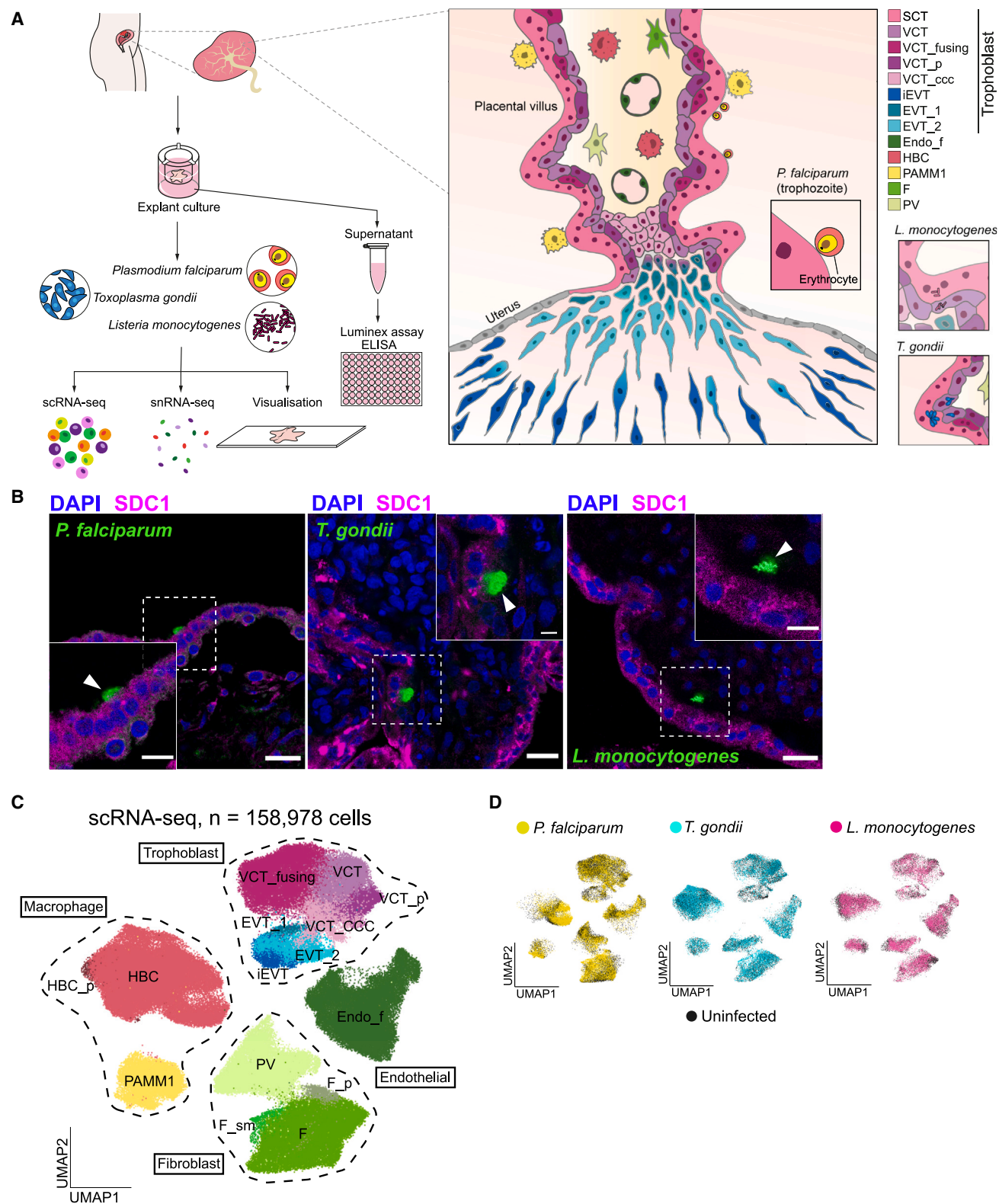


Figure 1. Infection of placental explants with multiple pathogens associated with intrauterine conditions

(A) (Left) Schematic of the workflow of the study. Fresh placental explants were dissected and grown in culture overnight and infected with *P. falciparum*, *T. gondii*, or *L. monocytogenes*. Samples were analyzed using single-cell RNA sequencing (scRNA-seq), single-nuclei RNA sequencing (snRNA-seq), high-resolution

(legend continued on next page)

recruiting immune cells to the infection site,^{26,27} and triggering tissue repair mechanisms^{28,29}), the function of HBCs in infection is still poorly characterized.

Recent advances to model placental infection *in vitro*³⁰ provide an alternative to mouse models, which exhibit substantial differences from the human placenta,^{31,32} including the location and origin of HBCs.^{33,34} Human trophoblast stem cells³⁵ and primary trophoblast organoids^{36,37} have recently been used to model Zika³⁸ and human cytomegalovirus³⁹ infection. In addition, placental explant cultures,^{13,40–44} where all placental lineages are represented, have been used to investigate infection burden to distinct pathogens. Bulk transcriptomics and profiling secreted immune factors from explants infected with *T. gondii* and *L. monocytogenes*^{13,44} have highlighted key inflammatory mechanisms in the placenta, yet the individual cell type responses and interactions across all immune and non-immune lineages remain poorly understood.

Here, we have generated the first comprehensive single-cell census of placental explants from early pregnancy infected with three pathogens associated with intrauterine complications, *L. monocytogenes* and *T. gondii*, which can cross the placenta, and *P. falciparum*, which cannot. When exposed to the three pathogens, all placental cells trigger inflammatory responses as an innate defense mechanism by 24 h post infection (hpi). In addition to the general activation of inflammatory pathways, each pathogen elicits a specific host response, which is most prominent in HBCs. Finally, we demonstrate that upon *P. falciparum* infection, the pathogen goes through molecular and metabolic adaptations, opening potential avenues to develop treatments against placental malaria.

RESULTS

Placental explants from early pregnancy maintain cellular identity and viability for more than 48 h

To study how human placental cells respond to infection during early pregnancy, a critical period when the risk of miscarriage after an infection is high,^{2,3,45} we optimized tissue explant cultures. We cultured 1 cm² placental explants between 5 and 14 post-conceptual weeks (PCWs) in an air-liquid interface,⁴⁶ using transwells coated with Matrigel, in the presence of modified trophoblast organoid media (TOM)³⁶ (Figure S1A; see STAR Methods). These culture conditions promoted visible outgrowth

of trophoblasts into the Matrigel by 48 h (Figure S1A), retained major hallmarks of trophoblast and macrophage identity (Figures S1B–S1D), and had high cell viability up to 72 h in culture (Figures S1E and S1F).

We profiled the explant model using single-cell RNA sequencing (scRNA-seq) and benchmarked it against our human *in vivo* placental cell atlas.⁴⁷ scRNA-seq of matched fresh tissue and placental explants demonstrated that our *ex vivo* culture system maintains all major cell lineages alive for at least 48 h, including trophoblasts (VCT, VCT_fusing), immune (HBCs), stromal (fibroblasts [F] and perivascular [PV]), and endothelial (Endo_f) (Figures S2A–S2C). In addition, we identified placenta-associated maternal macrophages (PAMM1), previously observed to be adhered to the outer surface of the SCT layer.^{17,48}

Major intrinsic innate immune signatures, including the expression of Toll-like receptors (TLRs) and interferon (IFN) receptors, are recapitulated in the macrophages and trophoblasts after 48 h of culture and are discernible even amid the transcriptomic changes associated with the culturing process (Figures S2D and S2E). Additional scRNA-seq on three distinct regions of the placenta from two independent samples demonstrated that cell types and proportions were consistent across different regions from the same donor, validating the robustness of the explant model (Figures S2F and S2G).

Taken together, our robust *ex vivo* placenta model can support all placental cell types in short-term cultures.

A single-cell census of human placental explants exposed to *P. falciparum*, *L. monocytogenes*, and *T. gondii*

We next exposed placental explants (~6–14 PCWs) to *P. falciparum*, *T. gondii*, and *L. monocytogenes* for 24 and 48 h. We chose the inocula for each microbe based on previous literature^{43,49,50} (see STAR Methods) and modified the culture media of the explants to maintain pathogen viability and infectious capacity (Figures 1A, 1B, and S3A; see STAR Methods). We enriched for *P. falciparum* parasites that express the receptor VAR2CSA, which encodes a *P. falciparum* erythrocyte membrane protein 1 (PfEMP1) that binds to chondroitin sulfate A (CSA)^{4,51} present in the SCT (Figures S3B and S3C; see STAR Methods). The enrichment of a VAR2CSA^{high} parasite population is essential, as continuous culturing of *P. falciparum* leads to heterogeneous *var* gene family expression, minimal

microscopy, and Luminex and ELISA immunoassays. (Right) Schematic of a placental villus embedded in the uterus showing all cell types and localization of different pathogens.

(B) Representative immunofluorescence whole-mount staining of infected placental explants 24 h post infection (hpi) in culture. Samples were stained with the SCT marker syndecan 1 (SDC1; magenta), anti-parasite/bacteria antibody (green; see Table S6), and DAPI (blue). (Left) *P. falciparum* (green, endogenous GFP expression) attached to the outside of the SCT; (middle) *T. gondii* tachyzoites (green) inside of the placental villus; (right) *L. monocytogenes* (green) inside of the VCT. Arrowheads indicate individual parasites (*P. falciparum* and *T. gondii*) or a group of bacteria (*L. monocytogenes*). Scale bars: main panels, 20 μ m; insets, 10 μ m.

(C) Uniform manifold approximation and projection (UMAP) of scRNA-seq ($n = 158,978$ cells; ArrayExpress: E-MTAB-12795) of uninfected and infected placental explants from matched donors 24 hpi ($n = 11$ donors; see Figure S4A) and 48 hpi ($n = 8$ donors; see Figure S4A). Cells are colored by cell state corresponding to the schematic right panel in (A).

(D) UMAPs of infected placental explants 24 and 48 hpi (ArrayExpress: E-MTAB-12795), showing the distribution of cells coming from both uninfected explants (black) and those infected explants with *P. falciparum* (yellow; $n = 4$ donors), *T. gondii* (cyan; $n = 4$ donors), and *L. monocytogenes* (magenta; $n = 3$ donors).

SCT, syncytiotrophoblast; VCT_fusing, villous cytotrophoblast fusing; VCT, villous cytotrophoblast; VCT_p, villous cytotrophoblast proliferative; VCT_CCC, villous cytotrophoblast cell column; iEVT, interstitial extravillous trophoblast; EVT_1, extravillous trophoblast 1; EVT_2, extravillous trophoblast 2; Endo_f, fetal endothelial; HBC, Hofbauer cell; HBC_p, Hofbauer cells proliferative; PAMM1, placenta-associated maternal macrophages; F, fibroblast; F_p, proliferative fibroblast; F_sm, fibroblast smooth muscle; PV, perivascular.

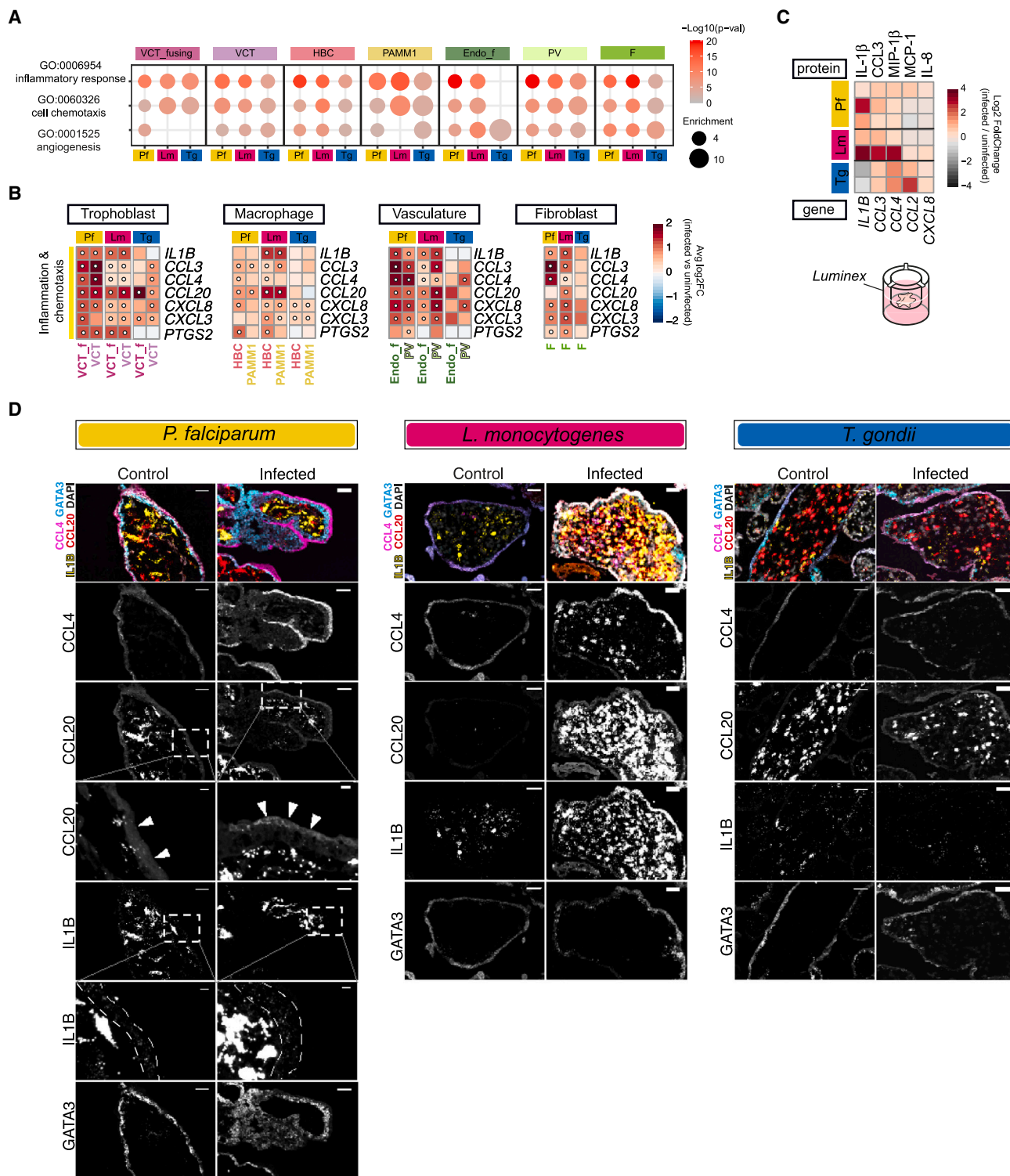


Figure 2. Induction of general inflammatory response in the placenta after infection with *P. falciparum*, *L. monocytogenes*, and *T. gondii* at 24 hpi

(A) Dot plot representing selected enriched Gene Ontologies (GO) for “inflammatory response,” “cell chemotaxis,” and “angiogenesis” for significantly up-regulated differentially expressed genes (DEGs) in the major placental lineages VCT_fusing, VCT, HBC, PAMM1, Endo_f, PV, and F. Comparisons were performed between uninfected and infected samples at 24 hpi across *P. falciparum* ($n = 4$ donors), *L. monocytogenes* ($n = 3$ donors), and *T. gondii* ($n = 4$ donors) infection (ArrayExpress: E-MTAB-12795). The size of the dot represents the enrichment score of the GO, and the color scale shows GO enrichment significance by $-\log_{10}(p \text{ value})$.

(legend continued on next page)

expression of *VAR2CSA*,⁵² and reduced infectious capacity (Figure S3D).

We then performed scRNA-seq to quantify placental responses during the early stages of infection at 24 and 48 hpi (Table S1) and identified all expected cell types in all samples (Figures 1A, 1C, 1D, S2A, and S4A–S4D). Our analysis confirmed the host cell expression of entry receptors/ligands for the three pathogens^{51,53–60} (Figure S4E), showing the suitability of our placental explants to study host responses to these three pathogens. In addition, VCT_fusing and VCT showed a higher number of differentially expressed genes (DEGs; adjusted $p < 0.05$) compared with extravillous trophoblasts (EVTs), the infiltrating trophoblasts in the uterus^{61–64} (Figure S4F). This result suggests that VCT_fusing and VCT are the primary cell types of the trophoblast compartment responding to the three pathogens. At 48 hpi, the transcriptional features associated with inflammation were less prominent (Figure S5A). Therefore, we chose to focus on the responses at 24 hpi.

Altogether, our placental explants serve as a robust multilineage system to effectively model the placenta's defense mechanisms against pathogens during the first 24 hpi.

General inflammatory response in the placenta after exposure to *P. falciparum*, *L. monocytogenes*, and *T. gondii*

We interrogated the responses of all placental lineages to the three pathogens at 24 hpi. We identified a cross-lineage enrichment in inflammatory responses, cellular chemotaxis, and angiogenesis programs (Figures 2A, 2B, and S5B). In the case of *P. falciparum* infection, this response is shared by trophoblasts and cells that cannot be directly infected, including fibroblasts, endothelial cells, and HBCs.

Cytokines and chemokines such as *CXCL3*, *CXCL8*, *CCL20*, *CCL4*, and *CCL3* were significantly upregulated by the three pathogens in trophoblasts (Figure 2B). This upregulation was also observed in the other placental lineages (immune, endothelial, and stroma), aside from *T. gondii* infection, where the increase did not pass the significance threshold (adjusted $p < 0.05$) in all these lineages (Figure 2B). Receptors for these cytokines and chemokines were expressed by multiple placental cell types, including HBCs and maternal PAMM1, and immune cells from the peripheral blood (Figures S5C and S5D). The fact that immune cells from both maternal and fetal compartments express receptors for the upregulated cytokines and chemokines suggests that both can be recruited to the site of infection in the placenta.

In comparison with the other pathogens, *T. gondii* infection resulted in a less widespread inflammatory and chemotaxis response across all major placental lineages, as shown by the lower p values in these two Gene Ontology (GO) categories that reflect a lower number of genes significantly upregulated (Figures 2A and 2B). One of the most distinctive features of *T. gondii* infection compared with the other infections was the absence of interleukin (*IL*)-1 β upregulation in the placental lineages (Figure 2B). *IL*-1 β is a key immunoregulatory and pro-inflammatory cytokine that is initially synthesized as an inactive precursor (pro-*IL*-1 β), which requires post-translational processing by caspase-1 upon inflammasome activation to become the active *IL*-1 β .⁶⁵ Our data showed that *T. gondii* could effectively circumvent this immune pathway in the placenta because neither increased transcription of *IL*-1 β nor its processing into pro-inflammatory *IL*-1 β at the protein level were observed (Figure 2C).

The increase in inflammatory cytokines observed in scRNA-seq data was confirmed at the tissue level using single-molecule fluorescent *in situ* hybridization (smFISH). In explants infected with *P. falciparum* and *L. monocytogenes*, we validated the upregulation of *CCL4* and *CCL20* across both the core of the placental villus and the GATA3+ trophoblasts (Figure 2D). In addition, scRNA-seq showed an upregulation of *CCL4* and *CCL20* exclusively in trophoblasts in response to *T. gondii*, and this was mirrored by the absence of increased signal in the placental core of the infected *T. gondii* explants (Figure 2D). Lastly, consistent with our scRNA-seq results, explants infected with *P. falciparum* and *L. monocytogenes* exhibited an increase in *IL*-1 β signals in both the placental core and trophoblasts, while *T. gondii*-infected explants did not show an increase in *IL*-1 β expression in any of the cell lineages (Figure 2D).

In sum, pathogen infection induced a general inflammatory response across all placental lineages, with very similar responses between *P. falciparum* and *L. monocytogenes*. *T. gondii* also upregulated inflammatory responses but to a lesser extent and without *IL*-1 β production.

Binding of *P. falciparum* leads to an altered FLT1/PGF ratio in trophoblasts

P. falciparum infection causes pregnancy complications by adhering to the SCT. We performed additional single-nuclei RNA sequencing (snRNA-seq) to detect the transcriptome of the multinucleated SCT in the presence of *P. falciparum* at 24 hpi (Figures S2A and S6A–S6D; Table S1). After exposure to *P. falciparum*, both the SCT (snRNA-seq) and VCT_fusing

** (B) Heatmap showing log₂ fold changes of inflammation and chemotaxis genes in major placental compartments, including trophoblast, macrophage, vasculature, and fibroblast. Comparisons were performed between uninfected and infected samples across *P. falciparum* ($n = 4$ donors), *L. monocytogenes* ($n = 3$ donors), and *T. gondii* ($n = 4$ donors) infection (ArrayExpress: E-MTAB-12795). Open white circles in the heatmap represent significance at false discovery rate (FDR) adjusted $p < 0.05$.

(C) Heatmap showing log₂ fold change of protein expression levels between uninfected and infected samples across multiple pathogens, detected through Luminex immunoassay of supernatant from explant cultures 24 hpi. Rows indicate biological replicates (Pf $n = 3$ donors, Lm $n = 2$ donors, and Tg $n = 2$ donors; see Figure S4A). Columns indicate protein and corresponding gene symbols.

(D) High-resolution large-area imaging by multiplexed single molecule fluorescent *in situ* hybridization (smFISH) of representative sections of *P. falciparum*, *L. monocytogenes*, and *T. gondii*-infected and matched uninfected placental explants at 24 hpi. Sections were stained with trophoblast marker GATA3 (cyan, white) and inflammatory markers *CCL4* (magenta, white), *IL*-1 β (yellow, white), and *CCL20* (red, white). Scale bars, 50 μ m and insets, 20 μ m.

White arrowheads—expression of *CCL20* in GATA3+ trophoblast layer. Dashed line demarcates the area of expression of *IL*-1 β in the trophoblast. Pf, *P. falciparum*; Lm, *L. monocytogenes*; Tg, *T. gondii*; VCT_fusing, villous cytotrophoblast fusing; VCT, villous cytotrophoblast; Endo_f, fetal endothelial; HBC, Hofbauer cell; PAMM1, placenta-associated maternal macrophages; F, fibroblast; PV, perivascular.

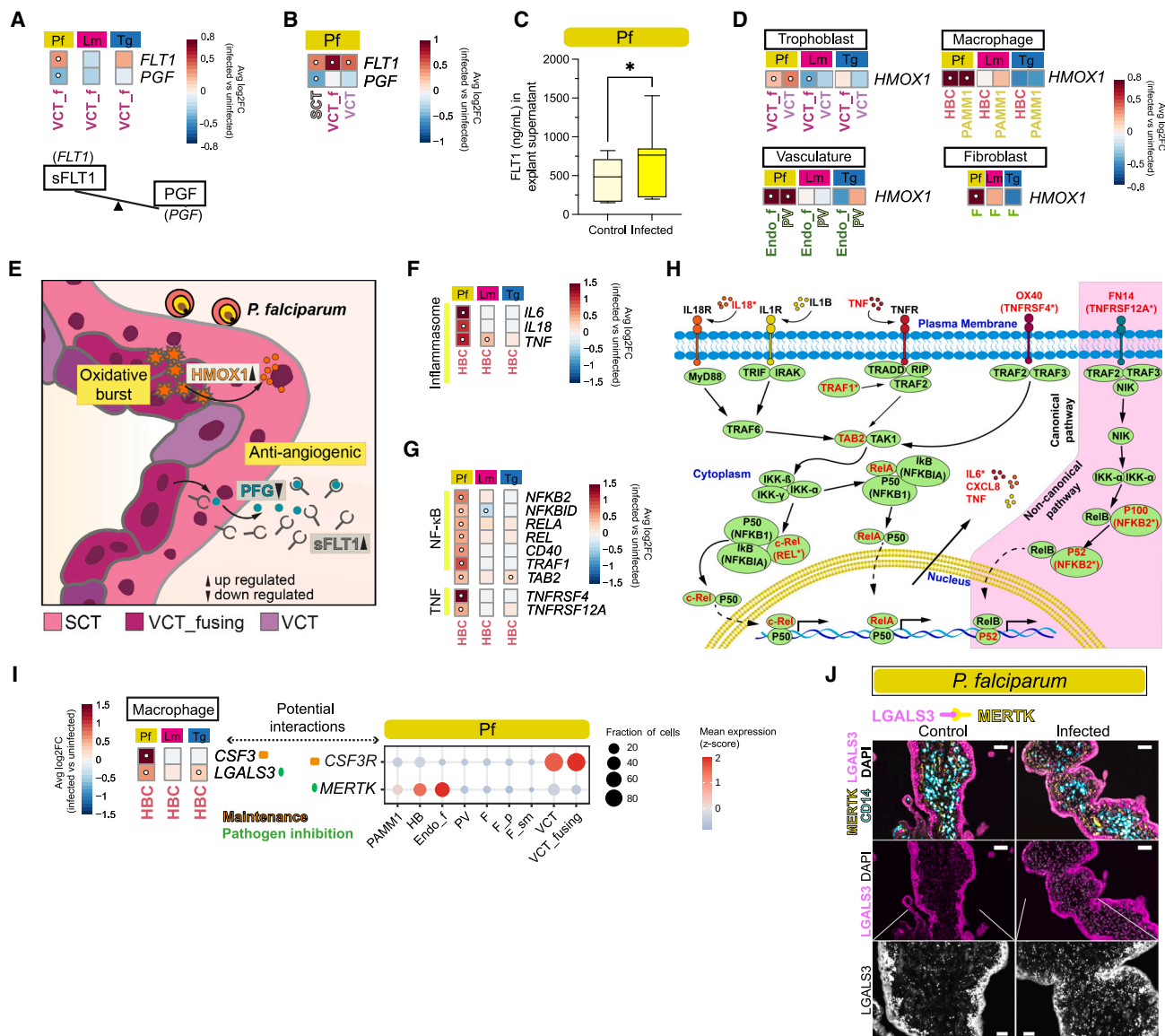


Figure 3. *P. falciparum* infection induces transcriptional changes in the trophoblast and promotes pro-inflammatory cytokines in HBCs

(A) Heatmap showing log₂ fold changes of *FLT1* and *PGF* in VCT_f at 24 hpi. Comparisons were performed between uninfected and infected samples across *P. falciparum* (*n* = 4 donors), *L. monocytogenes* (*n* = 3 donors), and *T. gondii* (*n* = 4 donors) infection (ArrayExpress: E-MTAB-12795). Open white circles in the heatmap represent significance at false discovery rate (FDR) adjusted *p* < 0.05.

(B) Heatmap showing log₂ fold changes of *FLT1* and *PGF* as measured by single-nuclei RNA sequencing (snRNA-seq) in SCT, VCT_f, and VCT at 24 hpi. Comparisons were performed between uninfected and infected samples across *P. falciparum* (*n* = 4 donors) infection (ArrayExpress: E-MTAB-12795). Open white circles in the heatmap represent significance at FDR adjusted *p* < 0.05.

(C) Boxplots showing amounts of sFLT1 in the supernatant of control and *P. falciparum*-exposed placental explants at 24 hpi, as determined by ELISA. Whiskers indicate minimum and maximum values. The lower edge of the box represents the 25th percentile, the line inside the box the median and the upper edge the 75th percentile. Asterisk (*) significance at *p* < 0.05 using Wilcoxon matched-pairs signed rank test, *n* = 7 donors.

(D) Heatmap showing log₂ fold changes of *HMOX1* at 24 hpi in major placental compartments, including trophoblast, macrophage, vasculature, and fibroblast. Comparisons were performed between uninfected and infected samples across *P. falciparum* (*n* = 4 donors), *L. monocytogenes* (*n* = 3 donors), and *T. gondii* (*n* = 4 donors) infection (ArrayExpress: E-MTAB-12795). Open white circles in the heatmap represent significance at FDR adjusted *p* < 0.05.

(E) Schematic representation of alterations in the trophoblast compartment triggered by the attachment of *P. falciparum* to the SCT.

(F and G) Heatmaps showing log₂ fold changes of inflammasome and NF-κB and TNF-related genes in HBC at 24 hpi. Comparisons were performed between uninfected and infected samples across *P. falciparum* (*n* = 4 donors), *L. monocytogenes* (*n* = 3 donors), and *T. gondii* (*n* = 4 donors) infection (ArrayExpress: E-MTAB-12795). Open white circles in the heatmap represent significance at FDR adjusted *p* < 0.05.

(H) Schematic representation of the NF-κB pathways in HBCs upon *P. falciparum* infection. Names in red represent genes that are significantly upregulated in HBCs upon *P. falciparum*. Asterisks indicate those genes that are only upregulated upon *P. falciparum*, but not after *L. monocytogenes* or *T. gondii* infections.

(legend continued on next page)

(precursor of SCT detected by scRNA-seq) downregulated the angiogenic gene placental growth factor (*PGF*)^{66,67} and upregulated its receptor fms related receptor tyrosin kinase 1 (*FLT1*) (Figures 3A and 3B). We used ELISA to confirm the significant increase of soluble FLT1 (sFLT1) in the infected explant culture supernatants (Figure 3C). Additionally, *P. falciparum* elicited upregulation of *HMOX1* in VCT_fusing and other placental cells, including macrophages, vasculature, and fibroblasts (Figure 3D). *HMOX1* is a key factor in mitigating malaria infections by controlling inflammation and protecting against heme-induced oxidative burst.^{68,69}

Thus, attachment of *P. falciparum* to the SCT resulted in the upregulation of factors that mitigate oxidative stress (*HMOX1*) and a change in expression of angiogenesis-related genes (*FLT1* and *PGF*), potentially disrupting normal placental angiogenesis (Figure 3E). A similarly altered sFLT1/PGF ratio is detected in pregnancy complications,⁷⁰ suggesting that *P. falciparum* promotes alterations in angiogenesis upon attachment to the SCT.

***P. falciparum* infection triggers expression of pro-inflammatory cytokines in HBCs**

Although *P. falciparum* does not cross the placenta^{4,71} (Figure 1B), it induced the upregulation of pro-inflammatory cytokines *IL-18*, tumor necrosis factor (*TNF*), and *IL-6* in HBCs (Figure 3F). The cognate receptors of these cytokines (*IL-6R*, *IL-18R1*, *TNFRSF1A*, *TNFRSF1B*, and *TNFRSF4*) were also expressed in HBCs (Figure S7A). Both *IL-18* and *TNF* are known activators of canonical nuclear factor (NF)- κ B^{72–75} and, accordingly, HBCs exposed to *P. falciparum* upregulated several members of the canonical NF- κ B pathway, including transcription factors (*REL* and *RELA*) and adaptors (*TRAF1* and *TAB2*) (Figures 3G and 3H). Moreover, HBCs from explants exposed to *P. falciparum* upregulated *TNFRSF12A*, the receptor for *TNFSF12*, which was also expressed in HBCs, fibroblasts, and endothelial cells (Figure S7B). In line with *TNFRSF12A* triggering the non-canonical NF- κ B pathway,^{76,77} HBCs exposed to *P. falciparum* upregulated *NFKB2*, a critical transcription factor of this pathway (Figures 3G and 3H).

HBCs from explants exposed to *P. falciparum* upregulated *CSF3* (granulocyte colony-stimulating factor [G-CSF]), a ligand for the *CSF3R* (GCSFR) receptor expressed by trophoblasts (Figure 3I). This receptor-ligand interaction has been previously demonstrated to support trophoblast development.^{78,79} HBCs from explants exposed to *P. falciparum* also upregulated galectin-3 (*LGALS3*), which is known to interact with MER proto-oncogene, tyrosine kinase (*MERTK*), a gene found ex-

pressed in HBCs (Figure 3I). *LGALS3-MERTK* interaction has been described to facilitate phagocytosis of apoptotic cells during inflammation,⁸⁰ pointing to its potential role in the placenta. We validated the expression of *MERTK* and the upregulation of *LGALS3* in HBCs using multiplexed smFISH (Figure 3J).

Taken together, our findings revealed that SCT-attached *P. falciparum* elicited an inflammatory response in HBCs, even though there is no direct contact between them.

Induction of antibacterial responses in HBCs upon *L. monocytogenes* infection

We next explored responses to *L. monocytogenes*, a pathogen able to infect various placental cell types in our explants (Figure S3A).^{2,6,13,20} Upon *L. monocytogenes* exposure, HBCs upregulated *TLR2*, a pathogen-recognition receptor crucial for activating the innate immune response against this bacteria (Figure 4A),^{81–83} *S100A8*, known to exacerbate inflammation in response to bacteria,^{84,85} and *SLC11A1*, *LYN*, and *HMGB1*, all of which have been shown to have antibacterial effects^{86–91} (Figure 4A). HBCs exposed to *L. monocytogenes* also upregulated *C5AR1*, a complement receptor described to protect against *L. monocytogenes* infection by preventing apoptosis of infected myeloid cells in a mouse model of listeriosis⁹² (Figure 4A).

In response to *L. monocytogenes*, HBCs upregulated the growth factor amphiregulin (*AREG*) (Figure 4B), an activator of the epidermal growth factor receptor (*EGFR*). This receptor-ligand interaction is known for promoting cell survival in trophoblasts.^{93–96} We validated the close proximity between *AREG* and *EGFR* by multiplexed smFISH (Figure 4C). HBCs upregulated *IL-1B*, while the components of the IL-1 receptor (*IL1R1* and *IL1RAP*) were expressed predominantly in fibroblasts and PV cells (Figure 4B), suggesting that *IL-1B* upregulation in HBCs mainly targets fibroblasts and vascular cells. HBCs also expressed IL-1 receptor components, albeit at lower levels (Figure 4B), hinting at an additional effect of *IL-1B* on these macrophages. Using multiplexed smFISH, we visualized the upregulation of *IL-1B* (Figure 4B) and the expression of the receptor *IL-1R1* (Figure 4D). This reinforces the notion that cells of the placental core were the primary targets of *IL-1B* upregulation.

In response to *L. monocytogenes* infection, HBCs therefore upregulated antimicrobial and pro-inflammatory molecules, as well as growth factors that may aid in renewal of trophoblasts.

***T. gondii* specifically targets HBCs in placental explants**

We next examined the responses to *T. gondii*, a highly motile parasite that actively infects host cells and establishes a niche within a vacuole that it populates and controls with its own

(I) (Left) Heatmap showing log₂ fold changes of *CSF3* and *LGALS3* in HBCs at 24 hpi. Comparisons were performed between uninfected and infected samples across *P. falciparum* (*n* = 4 donors), *L. monocytogenes* (*n* = 3 donors), and *T. gondii* (*n* = 4 donors) infection (ArrayExpress: E-MTAB-12795). Open white circles in the heatmap represent significance at FDR adjusted *p* < 0.05. (Right) Dot plots showing scaled Z score expression at 24 hpi of genes coding for potential interacting partners. The size of the dot represents the fraction (percentage) of cells expressing the gene. Specific interacting partners are linked with matching symbols, the color of the symbol indicates the biological role of the interaction listed below the heatmap. See Tables S3 and S4 for details about the individual interactions analyzed by CellphoneDBv4.

(J) High-resolution large-area imaging by single molecule fluorescent *in situ* hybridization (smFISH) of representative sections of *P. falciparum*-infected placental explants at 24 hpi. Sections stained with *LGALS3* (magenta), *MERTK* (yellow), macrophage marker *CD14* (cyan), and counterstained with DAPI to label nuclei. Scale bars, 50 μ m and insets, 20 μ m.

Pf, *P. falciparum*; Lm, *L. monocytogenes*; Tg, *T. gondii*; VCT_fusing, villous cytotrophoblast fusing; VCT, villous cytotrophoblast; VCT_p, villous cytotrophoblast proliferative; VCT_CCC, villous cytotrophoblast cell column; Endo_f, fetal endothelial; HBC, Hofbauer cell; PAMM1, placenta-associated maternal macrophages; F, fibroblast; F_p, proliferative fibroblast; F_sm, fibroblast smooth muscle; PV, perivascular.

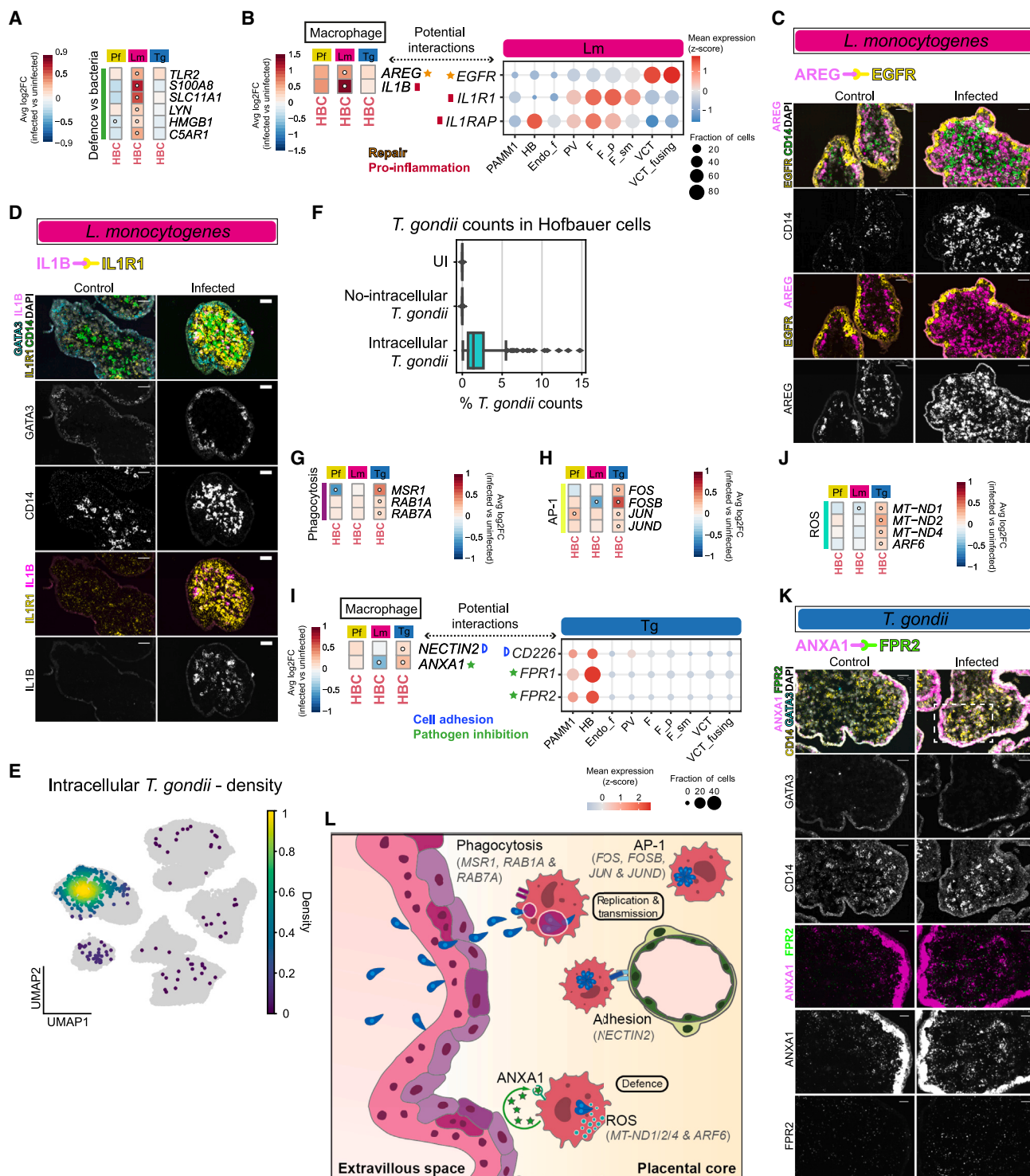


Figure 4. Antimicrobial role of fetal HBCs in the placental tissue

(A) Heatmap showing log₂ fold changes of bacterial-defense-related genes in HBC at 24 hpi. Comparisons were performed between uninfected and infected samples across *P. falciparum* (*n* = 4 donors), *L. monocytogenes* (*n* = 3 donors), and *T. gondii* (*n* = 4 donors) infection (ArrayExpress: E-MTAB-12795). Open white circles in the heatmap represent significance at false discovery rate (FDR) adjusted *p* < 0.05.

(B) (Left) Heatmap showing log₂ fold changes of *AREG* and *IL-1B* in HBC at 24 hpi. Comparisons were performed between uninfected and infected samples across *P. falciparum* (*n* = 4 donors), *L. monocytogenes* (*n* = 3 donors), and *T. gondii* (*n* = 4 donors) infection (ArrayExpress: E-MTAB-12795). Open white circles in the heatmap represent significance at FDR adjusted *p* < 0.05. (Right) Dot plots showing scaled Z score expression at 24 hpi of genes coding for potential interacting partners. The size of the dot represents the fraction (percentage) of cells expressing the gene. Specific interacting partners are linked with matching

(legend continued on next page)

proteins.⁹⁷ *T. gondii* can also hijack macrophages via phagocytosis, allowing internalization and subsequent escape from the phagosome compartment. This process involves deploying an invasive machinery to cross the phagosome membrane and settle within the macrophage.⁹⁸ Our scRNA-seq data successfully detected the polyadenylated (polyA) RNA of *T. gondii*, revealing that the majority of this signal is present in HBCs (Figures 4E, 4F, S7C, and S7D). This finding confirmed the presence of the parasite in the placental stromal core of the explants, where the HBCs are situated.

When exposed to *T. gondii*, HBCs upregulated phagocytosis-related genes, including *MSR1* and vesicular transport genes (*RAB1A* and *RAB7A*)^{99–102} (Figure 4G). This suggests that phagocytosis plays an important role in the enrichment of *T. gondii* in these cells. HBCs upregulated activator protein 1 (AP-1) transcription factors (*FOS*, *FOSB*, *JUN*, and *JUND*) (Figure 4H), the upregulation of which is required for the parasite's reproduction within its host,^{103–105} suggesting an additional mechanism by which *T. gondii* can exploit HBC intrinsic pathways. Moreover, HBCs upregulated the adhesion molecule *NECTIN2* in the presence of *T. gondii* (Figure 4I), the receptor of which, *CD226*,¹⁰⁶ was expressed by PV cells. This hints at a potential pathway for the recruitment of *T. gondii*-infected HBCs to the vasculature, which could lead to the parasite's spread to the fetal circulation.

In response to *T. gondii* infection, HBCs upregulated genes involved in the production of reactive oxygen species (ROS). This included subunits of nicotinamide adenine dinucleotide +

hydride (NADH) dehydrogenase (*MT-ND1*, *MT-ND2*, and *MT-ND4*), involved in the generation of superoxide (O_2^-), and *ARF6*, a small GTPase that controls nicotinamide adenine dinucleotide phosphate (NADPH) production and is responsible for ROS release (Figure 4J).^{107–109} The production of ROS is known to fight *T. gondii* in various contexts.^{107,108} In addition, HBCs upregulated annexin 1 (*ANXA1*) (Figure 4I), essential to control *T. gondii* load in the placenta,^{110,111} while the *ANXA1* receptors formyl peptide receptors 1 and 2 (*FPR1* and *FPR2*)¹¹² are predominantly expressed by HBCs (Figure 4I). The activation of *FPR1-2* is known to contribute to ROS production.^{113,114} We validated the expression of *FPR2* and the upregulation of *ANXA1* in HBCs using multiplexed smFISH (Figure 4K).

Altogether, we found that *T. gondii* was primarily concentrated in HBCs, where it exploited the host cells' intrinsic mechanisms (phagocytosis and AP-1) to aid its replication, potentially contributing to vertical transmission to the fetus (Figure 4L). Conversely, HBCs responded to *T. gondii* by generating ROS as a potential defense mechanism.

P. falciparum adapts to the placental microenvironment

We leveraged our explant model to study the adaptation of *P. falciparum* to the placental environment, taking advantage of the fact that the polyA RNA from malaria parasites can be detected by 10× Genomics.¹¹⁵ We performed scRNA-seq on *P. falciparum*-infected erythrocytes (2D9-VAR2CSA^{high} NF54 strain) that adhered to the SCT of placental explants after gentle dissociation¹¹⁶ (“pf_b,” binder). We then compared pf_b with

symbols, the color of the symbol indicates the biological role of the interaction listed below the heatmap. See Tables S3 and S4 for details about the individual interactions analyzed by CellPhoneDBv4.

(C) High-resolution large-area imaging by single molecule fluorescent *in situ* hybridization (smFISH) of representative sections of *L. monocytogenes*-infected placental explants at 24 hpi. Sections stained with *AREG* (magenta, white), *EGFR* (yellow), macrophage marker *CD14* (green, white), and counterstained with DAPI to label nuclei. Scale bars, 50 μ m.

(D) High-resolution large-area imaging by single smFISH of representative sections of *L. monocytogenes*-infected placental explants at 24 hpi. Sections stained with trophoblast marker *GATA3* (cyan, white), *IL-1B* (magenta, white), *IL-1R1* (yellow), macrophage marker *CD14* (green, white), and counterstained with DAPI to label nuclei. Scale bars, 50 μ m.

(E) Uniform manifold approximation and projection (UMAP) plot showing the density of cells with intracellular-*T. gondii* (intracellular-Tg) infected placental explants at 24 hpi in the single-cell embedding ($n = 4$ donors; ArrayExpress: E-MTAB-12795).

(F) Boxplot representing the percentage of *T. gondii* counts in HBCs at 24 hpi ($n = 4$ donors; ArrayExpress: E-MTAB-12795). Whiskers indicate minimum and maximum values. The lower edge of the box represents the 25th percentile, the line inside the box the median and the upper edge the 75th percentile. Uninfected samples (U); cells from infected samples without detectable intracellular *T. gondii* (no-intracellular-Tg); cells from infected samples with detectable intracellular *T. gondii* (intracellular-Tg).

(G and H) Heatmaps showing log₂ fold changes of phagocytosis-related and AP-1 transcription factors genes in HBC. Comparisons were performed between uninfected and infected samples across *P. falciparum* ($n = 4$ donors), *L. monocytogenes* ($n = 3$ donors), and *T. gondii* ($n = 4$ donors) infection (ArrayExpress: E-MTAB-12795). Open white circles in the heatmap represent significance at FDR adjusted $p < 0.05$.

(I) (Left) Heatmap showing log₂ fold changes of *NECTIN2* and *ANXA1* in HBCs at 24 hpi. Comparisons were performed between uninfected and infected samples across *P. falciparum* ($n = 4$ donors), *L. monocytogenes* ($n = 3$ donors), and *T. gondii* ($n = 4$ donors) infection (ArrayExpress: E-MTAB-12795). Open white circles in the heatmap represent significance at FDR adjusted $p < 0.05$. (Right) Dot plots showing scaled Z score expression of genes coding for potential interacting partners. The size of the dot represents the fraction (percentage) of cells expressing the gene. Specific interacting partners are linked with matching symbols, the color of the symbol indicates the biological role of the interaction listed below the heatmap. See Tables S3 and S4 for details about the individual interactions analyzed by CellPhoneDBv4.

(J) Heatmap showing log₂ fold changes of ROS-production-related genes in HBC at 24 hpi. Comparisons were performed between uninfected and infected samples across *P. falciparum* ($n = 4$ donors), *L. monocytogenes* ($n = 3$ donors), and *T. gondii* ($n = 4$ donors) infection (ArrayExpress: E-MTAB-12795). Open white circles in the heatmap represent significance at FDR adjusted $p < 0.05$.

(K) High-resolution large-area imaging by smFISH of representative sections of *T. gondii*-infected placental explants at 24 hpi. Sections stained with *ANXA1* (magenta, white), *FPR2* (green, white), macrophage marker *CD14* (yellow, white), trophoblast marker *GATA3* (cyan, white), and counterstained with DAPI to label nuclei. Scale bars, 50 μ m and insets, 20 μ m.

(L) Schematic representation of HBC responses upon 24 hpi *T. gondii* infection, encompassing processes involved in *T. gondii* replication and transmission and HBC antiparasitic defense.

Pf, *P. falciparum*; Lm, *L. monocytogenes*; Tg, *T. gondii*; VCT_fusing, villous cytotrophoblast fusing; VCT, villous cytotrophoblast; VCT_p, villous cytotrophoblast proliferative; VCT_CCC, villous cytotrophoblast cell column; Endo_f, fetal endothelial; HBC, Hofbauer cell; PAMM1, placenta-associated maternal macrophages; F, fibroblast; F_p, proliferative fibroblast; F_sm, fibroblast smooth muscle; PV, perivascular.

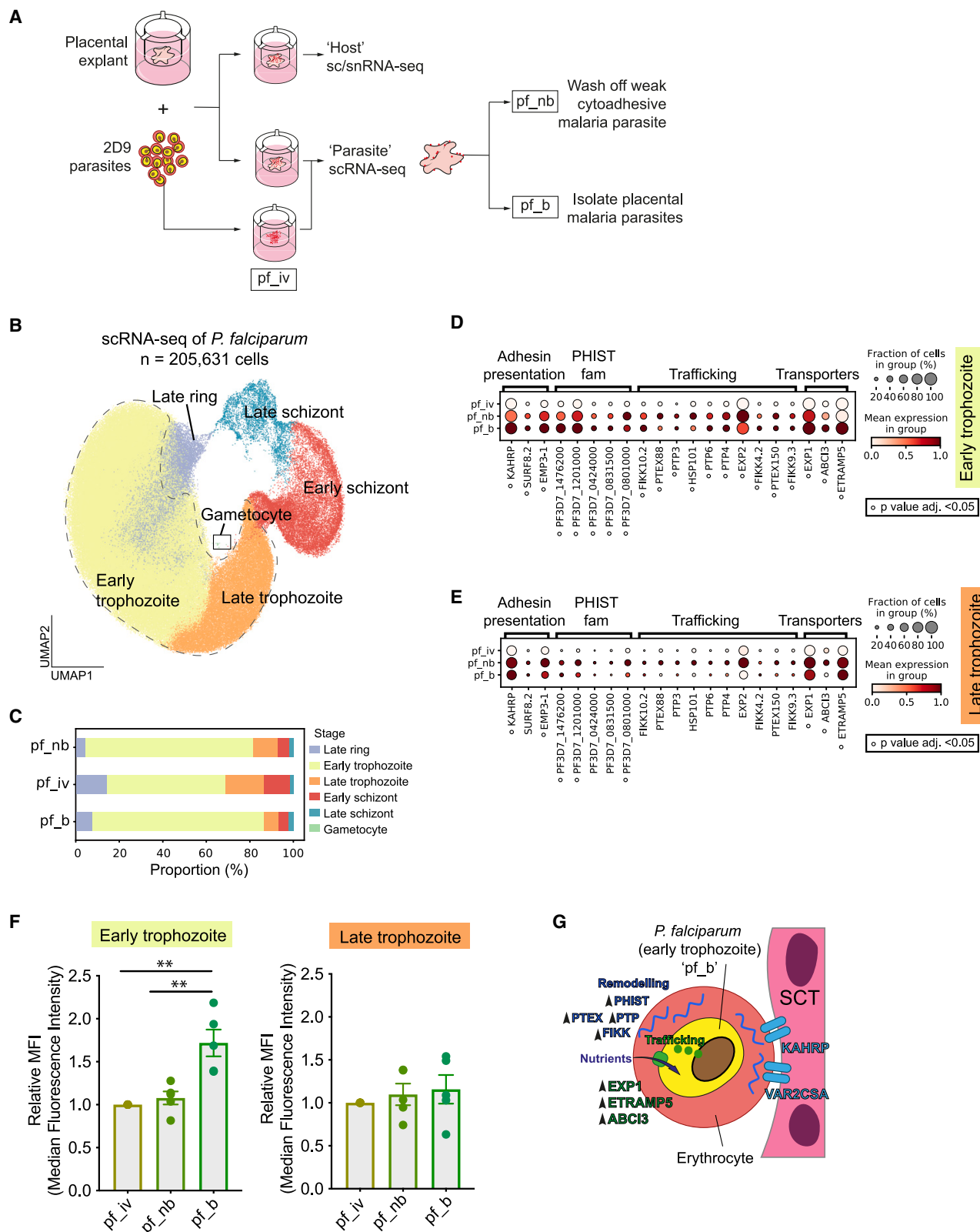


Figure 5. Placental infection alters adaptive programs in human malaria parasites

(A) Schematic of the workflow for *P. falciparum* culture and explant infections. pf_iv, pf_nb, and pf_b were analyzed using single-cell RNA sequencing (scRNA-seq).

(legend continued on next page)

P. falciparum-infected erythrocytes from the co-culture supernatant (“pf_nb,” non-binder fraction) as well as infected erythrocytes not previously exposed to the placenta (“pf_iv,” *in vitro*) (Figures 5A, S8A, and S8B; STAR Methods).

We profiled 205,631 *P. falciparum* single-cell transcriptomes from three independent placental explant infections (Figures 5B and S8C–S8E; Table S1). These experiments yielded a sufficient read depth (mean 10,000 reads/cell) to examine the local adaptation events of the malaria parasite upon its interaction with placental tissue (Table S2). The majority of the captured 2D9-VAR2CSA^{high} parasites were in early and late trophozoite stages, with a small proportion of early ring or dividing schizont stages (Figures 5B, 5C, and S8F). As expected, adherence to the placenta was strongest at the trophozoite stage, where VAR2CSA protein expression peaks.¹¹⁷

P. falciparum at the trophozoite stage extensively remodels its host erythrocyte compartment by inducing expression of surface adhesins, trafficking machinery and erythrocyte-exported proteins that allow infected erythrocytes to bind to host tissues, including the placenta.^{118–121} We found that in the early trophozoite stage, adherent parasites (pf_b) upregulated genes related to adhesin presentation (*KAHRP*, *SURF8.2*, and *EMP3*) and trafficking of PfEMP1 to the surface of host erythrocytes (*PTP3*, *PTP4*, *PTP6*,¹¹⁹ *FIKK10.2*, *FIKK9.3*, *FIKK4.2*,¹²² *EXP2*, *HSP101*, *PTEX88*, and *PTEX150*)¹²³ (Figure 5D). Additionally, members of the large family of exported *Plasmodium* helical interspersed subtelomeric (PHIST) proteins (*PF3D7_1476200*, *PF3D7_1201000*, *PF3D7_0424000*, *PF3D7_0831500*, and *PF3D7_0801000*)¹²⁴ were upregulated in pf_b (Figure 5D). Both adhesin-related and PHIST family members expressed by the parasites are known to regulate erythrocyte cytoskeleton remodeling. These genes were also upregulated in the late trophozoite stage, to a lesser extent (Figure 5E), as well as in the placenta-exposed non-adhered fraction (pf_nb) (Figure 5D). The non-adhered fraction included parasites that may have detached after being either weakly or transiently bound to the trophoblasts during the incubation period, potentially explaining this result.

Within the parasitophorous vacuolar compartment surrounding the malaria parasite, the parasite deploys nutrient-permeable channels and transporters that allow the influx of nutrients and small molecules.^{125–127} The adherent parasites (pf_b) in early trophozoites upregulated genes related to nutrient transport in the parasite (*EXP1*, *ABC13*, and *ETRAMP5*)¹²⁷ when compared with the non-exposed parasites (pf_iv) (Figure 5D). To validate

that binding of erythrocytes infected with *P. falciparum* to the placenta enhances nutrient uptake by the parasite, as suggested by our scRNA-seq data, we conducted a fluorescence-labeled glucose uptake assay comparing the pf_b adherent cells with non-binding controls (pf_nb and pf_iv). For glucose assay implementation and quantitative gating strategies, we first validated the glucose uptake assay in a previously described mutant Pf3D7 conditional EXP1 knockout line, complemented with EXP1wt^{low}-Ty1(sfa32)¹²⁷ (Figures S9A–S9E). We found that the adherent pf_b cells increased glucose uptake compared with pf_iv and pf_nb in the early trophozoite population (Figure 5F), supporting our hypothesis that binding to the trophoblast induces the transport of nutrients into the parasite. To rule out the possibility that changes in glucose uptake were caused by adherence to any surface via VAR2CSA, we performed the same experiment in the absence of placental explants. Binding to a chondroitin sulfate-coated petri dish was not sufficient to upregulate glucose uptake (Figure S9F), demonstrating that increased glucose uptake is specific to placenta adhesion. At the late trophozoite stage, there was no significant difference in glucose uptake between the parasite group pf_b versus pf_iv (Figure 5F), which may reflect the lack of significant expression of genes associated with erythrocyte cytoskeleton remodeling (*SURF8.2*, *PF3D7_0831500*, and *PF3D7_0424000*) and protein trafficking (*PTP3*, *PTP4*, *PTP6*, *FIKK10.2*, *FIKK9.3*, *FIKK4.2*, *EXP2*, *HSP101*, *PTEX88*, and *PTEX150*) (Figure 5E).

In conclusion, we found that *P. falciparum* adapted to the placental microenvironment by upregulating factors associated with erythrocyte cytoskeleton remodeling, adhesins and protein trafficking, and propose that *P. falciparum* can also adapt through modulation of nutrient uptake across the parasite membrane (Figure 5G).

DISCUSSION

Infections during pregnancy are globally common,^{128–132} yet the cellular responses in the placenta have been understudied compared with other barrier tissues.^{133–135} In this study, we investigated the acute response of early human placenta to three pathogens linked to pregnancy complications. We revealed a generalized inflammatory response across placental lineages against all three pathogens, independent of their route of infection, as well as pathogen-specific responses, particularly in HBCs. We also uncovered the strategies for *P. falciparum* to

(B) Uniform manifold approximation and projection (UMAP) of *P. falciparum* intraerythrocytic life cycle scRNA-seq ($n = 205,631$ cells; ArrayExpress: E-MTAB-12795) coming from pf_iv, pf_nb, and pf_b fractions.

(C) Bar plots showing proportion of *P. falciparum* intraerythrocytic life cycle stage coming from pf_iv ($n = 74,674$ cells), pf_nb ($n = 93,202$ cells), and pf_b ($n = 37,755$ cells) fractions (ArrayExpress: E-MTAB-12795).

(D) Dot plot of selected significant differentially expressed genes (DEGs) between pf_b and pf_iv at early trophozoite stage. Dot plots show normalized, log-transformed, and variance-scaled expression for early trophozoite pf_iv ($n = 40,780$ cells), pf_nb ($n = 71,625$ cells), and pf_b ($n = 29,846$ cells) fractions (ArrayExpress: E-MTAB-12795). Open circle adjacent to the gene name represents significance at false discovery rate (FDR) adjusted $p < 0.05$.

(E) Dot plot of selected significant DEGs between pf_b and pf_iv at late trophozoite stage. Dot plots show normalized, log-transformed, and variance-scaled expression for early trophozoite pf_iv ($n = 13,227$ cells), pf_nb ($n = 10,783$ cells), and pf_b ($n = 2,437$ cells) fractions (ArrayExpress: E-MTAB-12795). Open circle adjacent to the gene name represents significance at FDR adjusted $p < 0.05$.

(F) Relative median fluorescence intensity (MFI) of PE-glucose parasites coming from pf_iv, pf_nb, and pf_b fractions from placental infection. MFI values were normalized to pf_iv control to determine the relative change in pf_nb and pf_b parasites from early trophozoite (left) and late trophozoite (right) stage. ** $p < 0.01$ using non-parametric Mann-Whitney test, $n = 5$ independent experiments.

(G) Schematic diagram to illustrate pf_b and associated genes involved in adaptive change upon cytoadherence to human placental tissue.

SCT, syncytiotrophoblast; pf_iv, *P. falciparum* *in vitro* control; pf_nb, *P. falciparum* washed fraction with weak cytoadhesion; pf_b, *P. falciparum* placental binders.

adapt to the placental microenvironment. Our findings advance our understanding of the cellular and molecular responses in the human placenta upon pathogen exposure and introduce tools and resources for studying host-pathogen interactions across the fetal-maternal interface.

Regardless of whether the pathogen attached to the placenta (*P. falciparum*) or crossed it (*L. monocytogenes* and *T. gondii*), all lineages within the placenta swiftly activate inflammation programs upon infection. Although a local immune response is crucial for attempted pathogen clearance, uncontrolled production of inflammatory mediators can lead to systemic effects and hyperinflammation, resulting in adverse pregnancy outcomes.^{136,137} This is also observed during SARS-CoV-2 infection, where intense placental inflammation triggers the activation of maternal macrophages and causes defects in trophoblast differentiation, increasing the risk of pre-eclampsia, stillbirth, and preterm delivery.^{137–141} Various placental infections, including *P. falciparum*, *T. gondii*, chlamydia, and cytomegalovirus, have been associated with early onset pre-eclampsia.^{142–145} Placental expression of cytokines and other antimicrobial genes related to maternal infection are also associated with intervillitis,¹⁴⁶ a condition characterized by the infiltration of maternal immune cells into the placenta,¹⁴⁷ as occurs in placental malaria.¹⁴⁸ Additionally, higher levels of pro-inflammatory molecules have been linked to miscarriages,¹⁴⁹ spontaneous abortions, and preterm delivery.^{150–154}

Despite the many commonalities in how the placenta reacts to each of the studied infections, *T. gondii* uniquely, and unexpectedly, does not trigger inflammasome activation. Consequently, neither pro-IL-1 β upregulation nor secretion of its bioactive form into the explant supernatants are induced. The inhibition of the inflammasome has previously been observed in *T. gondii*-infected human neutrophils,¹⁵⁵ suggesting it might be a general evasion mechanism. In addition, *T. gondii* infection induces the NLRP3-dependent release of IL-1 β in adult human monocytes, but not in adult macrophages obtained from the same donor.¹⁵⁶ Collectively these data indicate a markedly cell- and context-specific regulation of the NLRP3 pathway and IL-1 β production by *T. gondii*, encompassing inflammasome suppression by HBCs.

P. falciparum has a tropism toward the trophoblast of the external placental layer (SCT). We found that *P. falciparum* induced an increased *sFLT1*/*PGF* ratio in SCT. An increased *sFLT1*/*PGF* ratio measured in plasma is associated with angiogenic defects in women with pregnancy complications^{70,157} and may result in altered fetal growth¹ and the systemic angiogenic changes seen in placental malaria.¹⁵⁸ The correlations between disruptions in placental core functions and pregnancy complications suggests that allocating placental resources to the inflammatory response during infection detracts from the direct embryo-supporting function.

Our study underscores the importance of HBCs in orchestrating the inflammatory response during the acute phase of infection. We demonstrated that HBCs also possess antimicrobial abilities in a physiological context and thus participate in the protection of the placenta. The study of their *in vivo* role in fighting infections within the placental context has been a major challenge and hence was poorly understood. Our placental explant setup with sustained viability of all cell

types presents a unique opportunity to interrogate these macrophages, which are derived from erythro-myeloid progenitors during the primitive wave of hematopoiesis, without a monocyte intermediate.^{18,34,159,160} In line with the plasticity of macrophage responses to distinct environments and insults,¹⁶¹ we found that HBCs trigger specific and characteristic programs in the presence of different types of pathogens. For example, the systemic inflammatory response caused by *P. falciparum* attachment to the SCT activates the NF- κ B pathway in HBCs, an indirect response that is likely mounted via trophoblast-produced inflammatory cytokines. Direct infection of *T. gondii* induced an AP-1-mediated response, while *L. monocytogenes* activated *IL-1B* and anti-microbial mechanisms. Our work uncovers, at high resolution, distinct pathogen-specific pathways activated in HBCs that could potentially inform the precision design of infection treatments.

HBCs were enriched with *T. gondii* reads, suggesting that HBCs are either targeted by *T. gondii*, engulf *T. gondii* tachyzoites or other infected cells, or experience a combination of these processes. Specific transcriptomic changes in *T. gondii*-infected HBCs revealed that *T. gondii* controls its host signaling pathways, likely to establish a permissive environment for parasite survival that can perpetuate infection. In turn, HBCs activated specific antimicrobial processes in response to *T. gondii* infection, such as acidification of cellular cytosol via ROS release, that would prevent parasite transmission to other cells. The fine balance between protective mechanisms from the HBCs and adaptive strategies deployed by the parasite is likely to determine infection outcome and HBC survival. We hypothesize that in cases where transmission does occur, *T. gondii* can turn infected macrophages into "Trojan horses" to disseminate the parasite across tissues.^{162,163} Supporting this idea, HBCs from infected explants upregulated ligands that could enhance their migration toward the placental vasculature, potentially leading to the transmission of *T. gondii* to the fetal circulation, as seen in Zika virus infection.^{164,165} The brain tropism shown by *T. gondii* may be due to the infection of circulating fetal microglia progenitors, similar to what is postulated in Zika virus infection.¹⁶⁶ Infected microglia progenitors may thus underlie the fetal death and distinctive brain malformations associated with *T. gondii*.³

Finally, by leveraging single-cell sequencing protocols for profiling polyA transcripts, we revealed aspects of both parasite and host during infection. *P. falciparum* is known to adapt to challenging host microenvironments such as the bone marrow, spleen, and liver^{167–169} to ensure its survival, replication, and transmission. We showed that malaria parasites interacted with the human placenta through SCT, and they increased adhesion, trafficking, and erythrocyte remodeling transcriptional programs, which facilitate cytoadherence and nutrient acquisition for parasite escape from host clearance and survival.^{170,171} These processes are known to play a crucial role in facilitating malaria parasite nutrient uptake from the external environment, growth, virulence, and host immune evasion during active infection.^{118,127,172,173} Notably, of the large family of PHIST exported proteins, two (*PF3D7_042400* and *PF3D7_120100*) that were upregulated in the parasites adhered to the SCT were also among the

significantly expressed genes in *P. falciparum* isolated from the placenta of pregnant Tanzanian women,¹⁷⁴ supporting the distinct adherent phenotypes for placental malaria parasites. Our deeper understanding of the erythrocyte remodeling and metabolic adaptive strategies employed by the parasite during placental malaria has potential implications for pathogenesis and the development of safe antimalarial treatments specifically designed for pregnant women in malaria endemic countries.

In conclusion, our study offers a panoramic view of infection-induced placental changes, along with an in-depth analysis of the interactions between host and pathogen cells during infection. We provide insights into the maternal-fetal interface, emphasizing that even the attachment of a pathogen to the trophoblast not only leads to inflammation but also has an impact on neighboring fetal immune cells and overall placental function. These challenges, along with dysregulation of cell-cell communication events upon infection, may disrupt the homeostatic function of the placenta. Our *in vitro* explant system is a tool to decode the placenta's defense strategies against infections, which we hope will be further harnessed and broadened to advance our understanding of infection during pregnancy, unlocking avenues for therapeutic intervention.

Limitations of the study

Our study represents a tailored approach to study the impact of infection on the early placenta. However, it only enables the study of early responses in both host and pathogen and does not allow for long-term infection monitoring or the identification of potential chronic biomarkers. Additionally, our explant model solely captures placental responses and does not account for potential interactions with other maternal immune cells found in the blood, such as natural killer (NK) cells and T cells, which are likely to be recruited to the infected SCT and play important roles in cellular defense. Due to the different infection behaviors of the pathogens used in this study, we were unable to estimate *in vivo* multiplicity of infection (MOI), the equivalent MOI across the pathogens per explant tissue, or control the number of bacteria or parasites actually infecting the tissue. Moreover, *P. falciparum* cannot progress to the next life cycle stage of reinventing erythrocytes at the 48 h time point in the explant culture and gaseous conditions. Thus, further refinement of the method is necessary to assess the placental responses to long-term infections and their impact on final pregnancy outcomes.

The placental explants used in this study are sourced from the Human Developmental Biology Resource. However, the detailed clinical status of the placental donors is unavailable. It is noteworthy that the placental donors did not exhibit symptoms of illness, such as fever, at the time of sample collection, suggesting that it is unlikely that they were infected with *P. falciparum*, *L. monocytogenes*, and *T. gondii* during this period. We reasoned that *P. falciparum* infection is not endemic in the United Kingdom (UK), with a relatively low number of travel-associated cases documented in individuals returning from malaria-prevalent regions.¹⁷⁵ This aligns with the limited prevalence of *T. gondii* infections (192 cases in 2022) and pregnancy-associated listeriosis cases (34 cases in 2021) reported in the UK.^{176,177}

Although droplet-based scRNA-seq can inform host cell-type-specific response, technical challenges impede the character-

ization of pathogens infecting the placenta at single-cell resolution. Gram-positive bacteria such as *L. monocytogenes* lack mRNA polyadenylation, and the low mRNA content for intracellular *T. gondii* among placental cells hinders the transcriptome profiling of these pathogens during the course of infection. Emerging technologies that can capture both host and pathogen transcripts could aid in addressing these limitations.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- RESOURCE AVAILABILITY
 - Lead contact
 - Materials availability
 - Data and code availability
- EXPERIMENTAL MODEL AND SUBJECT DETAILS
 - Samples
 - Placental explant culture
 - Culture of *L. monocytogenes*
 - Culture of *T. gondii* and tachyzoite isolation
 - Culture of human malaria parasite *Plasmodium falciparum* and *in vitro* selection procedures
- METHOD DETAILS
 - *var2csa* expression analysis by quantitative Real-Time PCR
 - Infection of human placental tissue explants
 - Processing of placental tissue explants for scRNA-seq
 - Isolation of *P. falciparum* 2D9-IEs from the placental tissue explants for scRNA-seq
 - Infection of CSA-coated petri dishes with *P. falciparum* 2D9-IEs
 - Processing of samples for snRNA-seq
 - 10x Genomics Chromium gene expression library preparation and sequencing
 - Flow cytometry
 - Tissue cryopreservation
 - Immunostaining and visualisation
 - Multiplexed smFISH and high-resolution imaging
 - Protein immunoassays
 - Glucose uptake assay
- QUANTIFICATION AND STATISTICAL ANALYSIS
 - Alignment and quantification of explant sc or snRNA-seq data, doublet detection and clustering
 - ScRNA-seq and snRNA-seq annotation and analysis of placental explants data
 - ScRNA-seq annotation and analysis of peripheral blood mononuclear cells (PBMCs)
 - Cell-cell communication predictions
 - Placenta hashing scRNA-seq analysis
 - *P. falciparum* scRNA-seq analysis
- ADDITIONAL RESOURCES

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.cels.2024.04.002>.

ACKNOWLEDGMENTS

This publication is part of the Human Cell Atlas. We thank Pascale Cossart (Pasteur Institute, Paris) for supplying the *L. monocytogenes* EGD-cGFP strain. We thank Sophie Adjalley for the PNF54eGFP strain. We thank Tobias Spielmann (Bernhard Nocht Institute for Tropical Medicine) for supplying the 3D7-EXP1wt^{low}-Ty1 (*sfa32*) condΔEXP1 line. We gratefully acknowledge the Sanger Cellular Generation and Phenotyping (CGaP) Core Facility and the Sanger Core Sequencing pipeline for support with sample processing

and sequencing library preparation; Bee Ling Ng and the Cytometry Core facility for their support and training; Tarryn Porter, Yvette Wood, Di Zhou, and the Cellular Genetics Wet Lab Support Team for their help in the wet lab logistics; David Goulding and the Advanced Light and Electron Microscopy suite; and Nick Thomson and Sally Kay for bacteria culture facilities. We thank Nicole Sevileno for her help in *P. falciparum* culture. We are grateful to Maximiliano Gutierrez and Amanda Oliver for their advice in metabolic dynamics and to Aidan Maartens and Valentina Lorenzi for proofreading. We thank Luz Garcia-Alonso for her advice in bioinformatics and Ana Paredes, Muzlifah Haniffa, and the Vento-Tormo lab for their comments on the manuscript. Placental material was provided by the Joint MRC-Human Cell Atlas (MR/S036350/1). The authors are grateful to patients for donating tissue for research. This research was funded by the Wellcome Trust grants 206194 and 220540/Z/20/A and the UK Research and Innovation (UKRI) under the UK government's Horizon Europe funding guarantee (grant number EP/Y009924/1). Additional support to R.F.W. was received from the Wellcome grant 214298/Z/18/Z.

AUTHOR CONTRIBUTIONS

R.V.-T. and R.H. conceived the study with contributions from E.R.R.-M.; R.H. and E.R.R.-M. set up and characterized the explant system; D.F.-A. provided scientific advice on the use of transwells and air-liquid interface culture for placental explants; *P. falciparum* cultures and glucose uptake experiments were performed by R.H. with contributions from M.R. and M.C.S.L.; *T. gondii* culture was done by E.R.R.-M. with contributions from S.C. and R.F.W.; *L. monocytogenes* cultures were done by R.H. and E.R.R.-M.; imaging experiments were performed by C.I.M., E.T., E.R.R.-M., R.H., and I.K.; infections and single-cell processing were done by E.R.R.-M., R.H., and I.K.; E.R.R.-M., R.H., and C.S.-S. performed single-nuclei extractions and library preparation; protein immunoassays were done by I.K., C.S.-S., and R.H.; data alignment and deconvolution pipelines were built by S.M.; E.R.R.-M. and R.H. analyzed and interpreted the data with contributions from R.V.-T., I.K., D.Á.-E., R.F.W., and M.C.S.L.; host-pathogen reference genomes were built by A.V.P. and E.R.R.-M.; E.R.R.-M. built and maintained the source code repositories; R.V.-T. and M.C.S.L. supervised the work; R.V.-T., E.R.R.-M., and R.H. wrote the manuscript with contributions from I.K., M.C.S.L., and D.Á.-E.; and the final version of the manuscript has been approved by all the authors.

DECLARATION OF INTERESTS

The authors declare no competing interests.

Received: May 19, 2023

Revised: December 1, 2023

Accepted: April 16, 2024

Published: May 3, 2024

REFERENCES

- Desai, M., ter Kuile, F.O., Nosten, F., McGready, R., Asamo, K., Brabin, B., and Newman, R.D. (2007). Epidemiology and burden of malaria in pregnancy. *Lancet Infect. Dis.* 7, 93–104. [https://doi.org/10.1016/S1473-3099\(07\)70021-X](https://doi.org/10.1016/S1473-3099(07)70021-X).
- Poulsen, K.P., and Czuprynski, C.J. (2013). Pathogenesis of listeriosis during pregnancy. *Anim. Health Res. Rev.* 14, 30–39. <https://doi.org/10.1017/S1466252312000242>.
- Arranz-Solís, D., Mukhopadhyay, D., and Saeij, J.J.P. (2021). Toxoplasma effectors that affect pregnancy outcome. *Trends Parasitol.* 37, 283–295. <https://doi.org/10.1016/j.pt.2020.10.013>.
- Fried, M., and Duffy, P.E. (1996). Adherence of *Plasmodium falciparum* to chondroitin sulfate A in the human placenta. *Science* 272, 1502–1504. <https://doi.org/10.1126/science.272.5267.1502>.
- Koppe, J.G., and Rothova, A. (1989). Congenital toxoplasmosis. A long-term follow-up of 20 years. *Int. Ophthalmol.* 13, 387–390. <https://doi.org/10.1007/BF02306486>.
- Topalovski, M., Yang, S.S., and Boonpasat, Y. (1993). Listeriosis of the placenta: clinicopathologic study of seven cases. *Am. J. Obstet. Gynecol.* 169, 616–620. [https://doi.org/10.1016/0002-9378\(93\)90632-s](https://doi.org/10.1016/0002-9378(93)90632-s).
- Maidji, E., McDonagh, S., Genbacev, O., Tabata, T., and Pereira, L. (2006). Maternal antibodies enhance or prevent cytomegalovirus infection in the placenta by neonatal Fc receptor-mediated transcytosis. *Am. J. Pathol.* 168, 1210–1226. <https://doi.org/10.2353/ajpath.2006.050482>.
- Lazar, M., Perelygina, L., Martinez, R., Greer, P., Paddock, C.D., Peltecu, G., Lupulescu, E., Icenogle, J., and Zaki, S.R. (2016). Immunolocalization and distribution of rubella antigen in fatal congenital rubella syndrome. *EBioMedicine* 3, 86–92.
- Bayer, A., Lennemann, N.J., Ouyang, Y., Bramley, J.C., Morosky, S., Marques, E.T.D.A., Jr., Cherry, S., Sadovsky, Y., and Coyne, C.B. (2016). Type III interferons Produced by Human Placental Trophoblasts Confer Protection against Zika Virus Infection. *Cell Host Microbe* 19, 705–712. <https://doi.org/10.1016/j.chom.2016.03.008>.
- Ganaie, S.S., and Qiu, J. (2018). Recent advances in replication and infection of human parvovirus B19. *Front. Cell. Infect. Microbiol.* 8, 166. <https://doi.org/10.3389/fcimb.2018.00166>.
- Park, J.-E., Jardine, L., Gottgens, B., Teichmann, S.A., and Haniffa, M. (2020). Prenatal development of human immunity. *Science* 368, 600–603. <https://doi.org/10.1126/science.aaz9330>.
- Hoo, R., Nakimuli, A., and Vento-Tormo, R. (2020). Innate immune mechanisms to protect against infection at the human decidual-placental interface. *Front. Immunol.* 11, 2070. <https://doi.org/10.3389/fimmu.2020.02070>.
- Megli, C., Morosky, S., Rajasundaram, D., and Coyne, C.B. (2021). Inflammasome signaling in human placental trophoblasts regulates immune defense against *Listeria monocytogenes* infection. *J. Exp. Med.* 218, e20200649. <https://doi.org/10.1084/jem.20200649>.
- Kwon, J.-Y., Aldo, P., You, Y., Ding, J., Racicot, K., Dong, X., Murphy, J., Glukshad, G., Silasi, M., Peng, J., et al. (2018). Relevance of placental type I interferon beta regulation for pregnancy success. *Cell. Mol. Immunol.* 15, 1010–1026. <https://doi.org/10.1038/s41423-018-0050-y>.
- Ding, J., Maxwell, A., Adzibolosu, N., Hu, A., You, Y., Liao, A., and Mor, G. (2022). Mechanisms of immune regulation by the placenta: role of type I interferon and interferon-stimulated genes signaling during pregnancy. *Immunol. Rev.* 308, 9–24. <https://doi.org/10.1111/imr.13077>.
- Kastschenko, N. (1885). Das menschliche Chorionepithel und dessen Rolle bei der Histogenese der Placenta. *Arch. Anat. Physiol.* 451, 480.
- Thomas, J.R., Appios, A., Zhao, X., Dutkiewicz, R., Donde, M., Lee, C.Y.C., Naidu, P., Lee, C., Cerveira, J., Liu, B., et al. (2021). Phenotypic and functional characterization of first-trimester human placental macrophages, Hofbauer cells. *J. Exp. Med.* 218, e20200891. <https://doi.org/10.1084/jem.20200891>.
- Thomas, J.R., Appios, A., Calderbank, E.F., Yoshida, N., Zhao, X., Hamilton, R.S., Moffett, A., Sharkey, A., Laurenti, E., Hanna, C.W., et al. (2023). Primitive haematopoiesis in the human placenta gives rise to macrophages with epigenetically silenced HLA-DR. *Nat. Commun.* 14, 1764. <https://doi.org/10.1038/s41467-023-37383-2>.
- Appios, A., Thomas, J.R., and McGovern, N. (2021). Isolation of first-trimester and full-term human placental Hofbauer cells. *Bio Protoc.* 11, e4044. <https://doi.org/10.21769/BioProtoc.4044>.
- Azari, S., Johnson, L.J., Webb, A., Kozlowski, S.M., Zhang, X., Rood, K., Amer, A., and Seveau, S. (2021). Hofbauer cells spread *Listeria monocytogenes* among placental cells and undergo pro-inflammatory reprogramming while retaining production of tolerogenic factors. *mBio* 12, e0184921. <https://doi.org/10.1128/mBio.01849-21>.
- Thomas, C.A., Li, Y., Kodama, T., Suzuki, H., Silverstein, S.C., and El Khoury, J. (2000). Protection from lethal gram-positive infection by macrophage scavenger receptor-dependent phagocytosis. *J. Exp. Med.* 191, 147–156. <https://doi.org/10.1084/jem.191.1.147>.

22. Brown, G.D., Taylor, P.R., Reid, D.M., Willment, J.A., Williams, D.L., Martinez-Pomares, L., Wong, S.Y.C., and Gordon, S. (2002). Dectin-1 is a major beta-glucan receptor on macrophages. *J. Exp. Med.* **196**, 407–412. <https://doi.org/10.1084/jem.20020470>.
23. Andrade, R.M., Wessendarp, M., Gubbels, M.-J., Striepen, B., and Subauste, C.S. (2006). CD40 induces macrophage anti-Toxoplasma gondii activity by triggering autophagy-dependent fusion of pathogen-containing vacuoles and lysosomes. *J. Clin. Invest.* **116**, 2366–2377. <https://doi.org/10.1172/JCI28796>.
24. Lian, Q., Zhang, K., Zhang, Z., Duan, F., Guo, L., Luo, W., Mok, B.W.-Y., Thakur, A., Ke, X., Motallebnejad, P., et al. (2022). Differential effects of macrophage subtypes on SARS-CoV-2 infection in a human pluripotent stem cell-derived model. *Nat. Commun.* **13**, 2028. <https://doi.org/10.1038/s41467-022-29731-5>.
25. Fujimoto, I., Pan, J., Takizawa, T., and Nakanishi, Y. (2000). Virus clearance through apoptosis-dependent phagocytosis of influenza A virus-infected cells by macrophages. *J. Virol.* **74**, 3399–3403. <https://doi.org/10.1128/jvi.74.7.3399-3403.2000>.
26. Van Furth, R., Diesselhoff-den Dulk, M.C., and Mattie, H. (1973). Quantitative study on the production and kinetics of mononuclear phagocytes during an acute inflammatory reaction. *J. Exp. Med.* **138**, 1314–1330. <https://doi.org/10.1084/jem.138.6.1314>.
27. Arango Duque, G., and Descoteaux, A. (2014). Macrophage cytokines: involvement in immunity and infectious diseases. *Front. Immunol.* **5**, 491. <https://doi.org/10.3389/fimmu.2014.00491>.
28. Varga, T., Mounier, R., Horvath, A., Cuvelier, S., Dumont, F., Poliska, S., Ardjoune, H., Juban, G., Nagy, L., and Chazaud, B. (2016). Highly dynamic transcriptional signature of distinct macrophage subsets during sterile inflammation, resolution, and tissue repair. *J. Immunol.* **196**, 4771–4782.
29. Mass, E., Nimmerjahn, F., Kierdorf, K., and Schlitzer, A. (2023). Tissue-specific macrophages: how they develop and choreograph tissue biology. *Nat. Rev. Immunol.* **23**, 563–579. <https://doi.org/10.1038/s41577-023-00848-y>.
30. Turco, M.Y., and Moffett, A. (2019). Development of the human placenta. *Development* **146**, dev163428. <https://doi.org/10.1242/dev.163428>.
31. Georgiades, P., Ferguson-Smith, A.C., and Burton, G.J. (2002). Comparative developmental anatomy of the murine and human definitive placenta. *Placenta* **23**, 3–19. <https://doi.org/10.1053/plac.2001.0738>.
32. Woods, L., Perez-Garcia, V., and Hemberger, M. (2018). Regulation of placental development and its impact on fetal growth—new insights from mouse models. *Front. Endocrinol.* **9**, 570. <https://doi.org/10.3389/fendo.2018.00570>.
33. Rhodes, K.E., Gekas, C., Wang, Y., Lux, C.T., Francis, C.S., Chan, D.N., Conway, S., Orkin, S.H., Yoder, M.C., and Mikkola, H.K.A. (2008). The emergence of hematopoietic stem cells is initiated in the placental vasculature in the absence of circulation. *Cell Stem Cell* **2**, 252–263. <https://doi.org/10.1016/j.stem.2008.01.001>.
34. Thomas, J.R., Naidu, P., Appios, A., and McGovern, N. (2021). The ontogeny and function of placental macrophages. *Front. Immunol.* **12**, 771054. <https://doi.org/10.3389/fimmu.2021.771054>.
35. Okae, H., Toh, H., Sato, T., Hiura, H., Takahashi, S., Shirane, K., Kabayama, Y., Suyama, M., Sasaki, H., and Arima, T. (2018). Derivation of human trophoblast stem cells. *Cell Stem Cell* **22**, 50–63.e6. <https://doi.org/10.1016/j.stem.2017.11.004>.
36. Turco, M.Y., Gardner, L., Kay, R.G., Hamilton, R.S., Prater, M., Hollinshead, M.S., McWhinnie, A., Esposito, L., Fernando, R., Skelton, H., et al. (2018). Trophoblast organoids as a model for maternal-fetal interactions during human placentation. *Nature* **564**, 263–267. <https://doi.org/10.1038/s41586-018-0753-3>.
37. Haider, S., Meinhardt, G., Saleh, L., Kunihs, V., Gamperl, M., Kaindl, U., Ellinger, A., Burkard, T.R., Fiala, C., Pollheimer, J., et al. (2018). Self-renewing trophoblast organoids recapitulate the developmental program of the early human placenta. *Stem Cell Rep.* **11**, 537–551. <https://doi.org/10.1016/j.stemcr.2018.07.004>.
38. Karvas, R.M., Khan, S.A., Verma, S., Yin, Y., Kulkarni, D., Dong, C., Park, K.-M., Chew, B., Sane, E., Fischer, L.A., et al. (2022). Stem-cell-derived trophoblast organoids model human placental development and susceptibility to emerging pathogens. *Cell Stem Cell* **29**, 810–825.e8. <https://doi.org/10.1016/j.stem.2022.04.004>.
39. Yang, L., Semmes, E.C., Ovies, C., Megli, C., Permar, S., Gilner, J.B., and Coyne, C.B. (2022). Innate immune signaling in trophoblast and decidua organoids defines differential antiviral defenses at the maternal-fetal interface. *eLife* **11**, e79794. <https://doi.org/10.7554/eLife.79794>.
40. Genbacev, O., Schubach, S.A., and Miller, R.K. (1992). Villous culture of first trimester human placenta—model to study extravillous trophoblast (EVT) differentiation. *Placenta* **13**, 439–461. [https://doi.org/10.1016/0143-4004\(92\)90051-t](https://doi.org/10.1016/0143-4004(92)90051-t).
41. Vicojac, L., Jones, C.J., and Aplin, J.D. (1995). Trophoblast differentiation during formation of anchoring villi in a model of the early human placenta in vitro. *Placenta* **16**, 41–56. [https://doi.org/10.1016/0143-4004\(95\)90080-2](https://doi.org/10.1016/0143-4004(95)90080-2).
42. Rizzuto, G.A., Kapidzic, M., Gormley, M., and Bakardjiev, A.I. (2016). Human placental and decidual organ cultures to study infections at the maternal-fetal interface. *J. Vis. Exp.* <https://doi.org/10.3791/54237>.
43. Crespo, A.C., Mulik, S., Dotiwala, F., Ansara, J.A., Sen Santara, S., Ingersoll, K., Ovies, C., Junqueira, C., Tilburgs, T., Strominger, J.L., et al. (2020). Decidual NK cells transfer granulysin to selectively kill bacteria in trophoblasts. *Cell* **182**, 1125–1139.e18. <https://doi.org/10.1016/j.cell.2020.07.019>.
44. Ander, S.E., Rudzki, E.N., Arora, N., Sadovsky, Y., Coyne, C.B., and Boyle, J.P. (2018). Human placental syncytiotrophoblasts restrict Toxoplasma gondii attachment and replication and respond to infection by producing immunomodulatory chemokines. *mBio* **9**, e01678–e016717. <https://doi.org/10.1128/mBio.01678-17>.
45. Mylonakis, E., Paliou, M., Hohmann, E.L., Calderwood, S.B., and Wing, E.J. (2002). Listeriosis during pregnancy: a case series and review of 222 cases. *Medicine* **81**, 260–269. <https://doi.org/10.1097/00005792-200207000-00002>.
46. Herms, A., Fernandez-Antoran, D., Alcolea, M.P., Kalogeropoulou, A., Banerjee, U., Piedrafita, G., Abby, E., Valverde-Lopez, J.A., Ferreira, I.S., Dentro, S.C., et al. (2023). Epithelioids: Self-sustaining 3D epithelial cultures to study long-term processes. Preprint at bioRxiv. <https://doi.org/10.1101/2023.01.03.522589>.
47. Arutyunyan, A., Roberts, K., Troulé, K., Wong, F.C.K., Sheridan, M.A., Kats, I., Garcia-Alonso, L., Velten, B., Hoo, R., Ruiz-Morales, E.R., et al. (2023). Spatial multiomics map of trophoblast development in early pregnancy. *Nature* **616**, 143–151. <https://doi.org/10.1038/s41586-023-05869-0>.
48. Vento-Tormo, R., Efremova, M., Botting, R.A., Turco, M.Y., Vento-Tormo, M., Meyer, K.B., Park, J.-E., Stephenson, E., Polański, K., Goncalves, A., et al. (2018). Single-cell reconstruction of the early maternal-fetal interface in humans. *Nature* **563**, 347–353. <https://doi.org/10.1038/s41586-018-0698-6>.
49. World Health Organization (2012). *Management of Severe Malaria: A Practical Handbook* (WHO).
50. Robbins, J.R., Zeldovich, V.B., Poukchanski, A., Boothroyd, J.C., and Bakardjiev, A.I. (2012). Tissue barriers of the human placenta to infection with Toxoplasma gondii. *Infect. Immun.* **80**, 418–428. <https://doi.org/10.1128/IAI.05899-11>.
51. Salanti, A., Dahlbäck, M., Turner, L., Nielsen, M.A., Barfod, L., Magistrado, P., Jensen, A.T.R., Lavstsen, T., Ofori, M.F., Marsh, K., et al. (2004). Evidence for the involvement of VAR2CSA in pregnancy-associated malaria. *J. Exp. Med.* **200**, 1197–1203. <https://doi.org/10.1084/jem.20041579>.
52. Frank, M., Dzikiowski, R., Amulic, B., and Deitsch, K. (2007). Variable switching rates of malaria virulence genes are associated with chromosomal position. *Mol. Microbiol.* **64**, 1486–1498. <https://doi.org/10.1111/j.1365-2958.2007.05736.x>.

53. Lecuit, M., Nelson, D.M., Smith, S.D., Khun, H., Huerre, M., Vacher-Lavenu, M.-C., Gordon, J.I., and Cossart, P. (2004). Targeting and crossing of the human maternofetal barrier by *Listeria monocytogenes*: role of internalin interaction with trophoblast E-cadherin. *Proc. Natl. Acad. Sci. USA* **101**, 6152–6157. <https://doi.org/10.1073/pnas.0401434101>.
54. Pizarro-Cerdá, J., and Cossart, P. (2018). *Listeria monocytogenes*: cell biology of invasion and intracellular growth. *Microbiol. Spectr.* **6**, 10–1128. <https://doi.org/10.1128/microbiolspec.GPP3-0013-2018>.
55. Bonhomme, A., Bouchot, A., Pezzella, N., Gomez, J., Le Moal, H., and Pinon, J.M. (1999). Signaling during the invasion of host cells by *Toxoplasma gondii*. *FEMS Microbiol. Rev.* **23**, 551–561. <https://doi.org/10.1111/j.1574-6976.1999.tb00413.x>.
56. Furtado, G.C., Cao, Y., and Joiner, K.A. (1992). Laminin on *Toxoplasma gondii* mediates parasite binding to the beta 1 integrin receptor alpha 6 beta 1 on human foreskin fibroblasts and Chinese hamster ovary cells. *Infect. Immun.* **60**, 4925–4931. <https://doi.org/10.1128/iai.60.11.4925-4931.1992>.
57. Barragan, A., Brossier, F., and Sibley, L.D. (2005). Transepithelial migration of *Toxoplasma gondii* involves an interaction of intercellular adhesion molecule 1 (ICAM-1) with the parasite adhesin MIC2. *Cell. Microbiol.* **7**, 561–568. <https://doi.org/10.1111/j.1462-5822.2005.00486.x>.
58. Nasuhidehnavi, A., Zhao, Y., Punetha, A., Hemphill, A., Li, H., Bechtel, T.J., Rager, T., Xiong, B., Petrou, V.I., Gubbels, M.-J., et al. (2022). A role for basigin in *Toxoplasma gondii* infection. *Infect. Immun.* **90**, e0020522. <https://doi.org/10.1128/iai.00205-22>.
59. Salanti, A., Clausen, T.M., Agerbæk, M.Ø., Al Nakouzi, N., Dahlbäck, M., Oo, H.Z., Lee, S., Gustavsson, T., Rich, J.R., Hedberg, B.J., et al. (2015). Targeting human cancer by a glycosaminoglycan binding malaria protein. *Cancer Cell* **28**, 500–514. <https://doi.org/10.1016/j.ccell.2015.09.003>.
60. Ayres Pereira, M., Mandel Clausen, T., Pehrson, C., Mao, Y., Resende, M., Daugaard, M., Riis Kristensen, A., Spliid, C., Mathiesen, L., Knudsen, L.E., et al. (2016). Placental sequestration of *Plasmodium falciparum* malaria parasites is mediated by the interaction between VAR2CSA and chondroitin sulfate A on Syndecan-1. *PLoS Pathog.* **12**, e1005831. <https://doi.org/10.1371/journal.ppat.1005831>.
61. Benirschke, K., Burton, G.J., and Baergen, R.N. (2012). *Pathology of the Human Placenta* (Springer).
62. Burton, G.J., and Jauniaux, E. (2017). The cytotrophoblastic shell and complications of pregnancy. *Placenta* **60**, 134–139. <https://doi.org/10.1016/j.placenta.2017.06.007>.
63. Jauniaux, E., Watson, A.L., Hempstock, J., Bao, Y.P., Skepper, J.N., and Burton, G.J. (2000). Onset of maternal arterial blood flow and placental oxidative stress. A possible factor in human early pregnancy failure. *Am. J. Pathol.* **157**, 2111–2122. [https://doi.org/10.1016/S0002-9440\(10\)64849-3](https://doi.org/10.1016/S0002-9440(10)64849-3).
64. Burton, G.J., Woods, A.W., Jauniaux, E., and Kingdom, J.C.P. (2009). Rheological and physiological consequences of conversion of the maternal spiral arteries for uteroplacental blood flow during human pregnancy. *Placenta* **30**, 473–482. <https://doi.org/10.1016/j.placenta.2009.02.009>.
65. Martinon, F., Burns, K., and Tschopp, J. (2002). The inflammasome: a molecular platform triggering activation of inflammatory caspases and processing of proIL-beta. *Mol. Cell* **10**, 417–426. [https://doi.org/10.1016/S1097-2765\(02\)00599-3](https://doi.org/10.1016/S1097-2765(02)00599-3).
66. Clauss, M., Weich, H., Breier, G., Knies, U., Röckl, W., Waltenberger, J., and Risau, W. (1996). The vascular endothelial growth factor receptor Flt-1 mediates biological activities: Implications for a functional role of placenta growth factor in monocyte activation and chemotaxis. *J. Biol. Chem.* **271**, 17629–17634. <https://doi.org/10.1074/jbc.271.30.17629>.
67. Maynard, S.E., Min, J.-Y., Merchan, J., Lim, K.-H., Li, J., Mondal, S., Libermann, T.A., Morgan, J.P., Sellke, F.W., Stillman, I.E., et al. (2003). Excess placental soluble fms-like tyrosine kinase 1 (sFlt1) may contribute to endothelial dysfunction, hypertension, and proteinuria in preeclampsia. *J. Clin. Invest.* **111**, 649–658. <https://doi.org/10.1172/JCI117189>.
68. Pamplona, A., Ferreira, A., Balla, J., Jeney, V., Balla, G., Epiphanio, S., Chora, A., Rodrigues, C.D., Gregoire, I.P., Cunha-Rodrigues, M., et al. (2007). Heme oxygenase-1 and carbon monoxide suppress the pathogenesis of experimental cerebral malaria. *Nat. Med.* **13**, 703–710. <https://doi.org/10.1038/nm1586>.
69. Seixas, E., Gozzelino, R., Chora, A., Ferreira, A., Silva, G., Larsen, R., Rebelo, S., Penido, C., Smith, N.R., Coutinho, A., et al. (2009). Heme oxygenase-1 affords protection against noncerebral forms of severe malaria. *Proc. Natl. Acad. Sci. USA* **106**, 15837–15842. <https://doi.org/10.1073/pnas.0903419106>.
70. Zeisler, H., Llurba, E., Chantraine, F., Vathis, M., Staff, A.C., Sennström, M., Olovsson, M., Brennecke, S.P., Stepan, H., Allegranza, D., et al. (2016). Predictive value of the sFlt-1:PlGF ratio in women with suspected preeclampsia. *N. Engl. J. Med.* **374**, 13–22. <https://doi.org/10.1056/NEJMoa1414838>.
71. Muthusamy, A., Achur, R.N., Bhavanandan, V.P., Fouda, G.G., Taylor, D.W., and Gowda, D.C. (2004). *Plasmodium falciparum*-infected erythrocytes adhere both in the intervillous space and on the villous surface of human placenta by binding to the low-sulfated chondroitin sulfate proteoglycan receptor. *Am. J. Pathol.* **164**, 2013–2025. [https://doi.org/10.1016/S0002-9440\(10\)63761-3](https://doi.org/10.1016/S0002-9440(10)63761-3).
72. Kojima, H., Takeuchi, M., Ohta, T., Nishida, Y., Arai, N., Ikeda, M., Ikegami, H., and Kurimoto, M. (1998). Interleukin-18 activates the IRAK-TRAF6 pathway in mouse EL-4 cells. *Biochem. Biophys. Res. Commun.* **244**, 183–186. <https://doi.org/10.1006/bbrc.1998.8236>.
73. Yasuda, K., Nakanishi, K., and Tsutsui, H. (2019). Interleukin-18 in Health and Disease. *Int. J. Mol. Sci.* **20**, 649. <https://doi.org/10.3390/ijms20030649>.
74. Liu, T., Zhang, L., Joo, D., and Sun, S.-C. (2017). NF-κB signaling in inflammation. *Signal Transduct. Target. Ther.* **2**, 17023. <https://doi.org/10.1038/sigtrans.2017.23>.
75. Yu, H., Lin, L., Zhang, Z., Zhang, H., and Hu, H. (2020). Targeting NF-κB pathway for the therapy of diseases: mechanism and clinical study. *Signal Transduct. Target. Ther.* **5**, 209. <https://doi.org/10.1038/s41392-020-00312-6>.
76. Saitoh, T., Nakayama, M., Nakano, H., Yagita, H., Yamamoto, N., and Yamaoka, S. (2003). TWEAK induces NF-κB2 p100 processing and long lasting NF-κB activation. *J. Biol. Chem.* **278**, 36005–36012. <https://doi.org/10.1074/jbc.M304266200>.
77. Sun, S.-C. (2017). The non-canonical NF-κB pathway in immunity and inflammation. *Nat. Rev. Immunol.* **17**, 545–558. <https://doi.org/10.1038/nri.2017.52>.
78. Furmento, V.A., Marino, J., Blank, V.C., and Roguin, L.P. (2014). The granulocyte colony-stimulating factor (G-CSF) upregulates metalloproteinase-2 and VEGF through PI3K/Akt and ERK1/2 activation in human trophoblast Swan 71 cells. *Placenta* **35**, 937–946. <https://doi.org/10.1016/j.placenta.2014.09.003>.
79. Gao, P., Zha, Y., Wei, L., Zhou, X., Zhu, S., Zhang, H., Gao, X., Jiang, Y., Chen, Y., Li, J., et al. (2022). G-CSF: A vehicle for communication between trophoblasts and macrophages which may cause problems in recurrent spontaneous abortion. *Placenta* **121**, 164–172. <https://doi.org/10.1016/j.placenta.2022.03.125>.
80. Caberoy, N.B., Alvarado, G., Bigcas, J.-L., and Li, W. (2012). Galectin-3 is a new MerTK-specific eat-me signal. *J. Cell. Physiol.* **227**, 401–407. <https://doi.org/10.1002/jcp.22955>.
81. Flo, T.H., Halaas, O., Lien, E., Ryan, L., Teti, G., Golenbock, D.T., Sundan, A., and Espevik, T. (2000). Human toll-like receptor 2 mediates monocyte activation by *Listeria monocytogenes*, but not by group B streptococci or lipopolysaccharide. *J. Immunol.* **164**, 2064–2069. <https://doi.org/10.4049/jimmunol.164.4.2064>.
82. Torres, D., Barrier, M., Bihl, F., Quesniaux, V.J.F., Maillet, I., Akira, S., Ryffel, B., and Erard, F. (2004). Toll-like receptor 2 is required for optimal control of *Listeria monocytogenes* infection. *Infect. Immun.* **72**, 2131–2139. <https://doi.org/10.1128/IAI.72.4.2131-2139.2004>.

83. Janot, L., Secher, T., Torres, D., Maillet, I., Pfeilschifter, J., Quesniaux, V.F.J., Landmann, R., Ryffel, B., and Erard, F. (2008). CD14 works with toll-like receptor 2 to contribute to recognition and control of *Listeria monocytogenes* infection. *J. Infect. Dis.* 198, 115–124. <https://doi.org/10.1086/588815>.
84. Nacken, W., Roth, J., Sorg, C., and Kerkhoff, C. (2003). S100A9/S100A8: myeloid representatives of the S100 protein family as prominent players in innate immunity. *Microsc. Res. Tech.* 60, 569–580. <https://doi.org/10.1002/jemt.10299>.
85. Wang, S., Song, R., Wang, Z., Jing, Z., Wang, S., and Ma, J. (2018). S100A8/A9 in Inflammation. *Front. Immunol.* 9, 1298. <https://doi.org/10.3389/fimmu.2018.01298>.
86. Yanai, H., Matsuda, A., An, J., Koshiba, R., Nishio, J., Negishi, H., Ikushima, H., Onoe, T., Ohdan, H., Yoshida, N., et al. (2013). Conditional ablation of HMGB1 in mice reveals its protective function against endotoxemia and bacterial infection. *Proc. Natl. Acad. Sci. USA* 110, 20699–20704. <https://doi.org/10.1073/pnas.1320808110>.
87. Canonne-Hergaux, F., Gruenheid, S., Govoni, G., and Gros, P. (1999). The Nrp1 protein and its role in resistance to infection and macrophage function. *Proc. Assoc. Am. Phys.* 111, 283–289. <https://doi.org/10.1046/j.1525-1381.1999.99236.x>.
88. Brown, D.E., Libby, S.J., Moreland, S.M., McCoy, M.W., Brabb, T., Stepanek, A., Fang, F.C., and Detweiler, C.S. (2013). *Salmonella enterica* causes more severe inflammatory disease in C57/BL6 Nrp1G169 mice than Sv129S6 mice. *Vet. Pathol.* 50, 867–876. <https://doi.org/10.1177/0300985813478213>.
89. Cunrath, O., and Bumann, D. (2019). Host resistance factor SLC11A1 restricts *Salmonella* growth through magnesium deprivation. *Science* 366, 995–999. <https://doi.org/10.1126/science.aax7898>.
90. Li, X., He, S., Zhou, X., Ye, Y., Tan, S., Zhang, S., Li, R., Yu, M., Jundt, M.C., Hidebrand, A., et al. (2016). Lyn delivers bacteria to lysosomes for eradication through TLR2-initiated autophagy related phagocytosis. *PLoS Pathog.* 12, e1005363. <https://doi.org/10.1371/journal.ppat.1005363>.
91. Li, X., Zhou, X., Ye, Y., Li, Y., Li, J., Privratsky, B., Wu, E., Gao, H., Huang, C., and Wu, M. (2014). Lyn regulates inflammatory responses in *Klebsiella pneumoniae* infection via the p38/NF- κ B pathway. *Eur. J. Immunol.* 44, 763–773. <https://doi.org/10.1002/eji.201343972>.
92. Calame, D.G., Mueller-Ortiz, S.L., Morales, J.E., and Wetsel, R.A. (2014). The C5a anaphylatoxin receptor (C5aR1) protects against *Listeria monocytogenes* infection by inhibiting type 1 IFN expression. *J. Immunol.* 193, 5099–5107. <https://doi.org/10.4049/jimmunol.1401750>.
93. Sanderson, M.P., Dempsey, P.J., and Dunbar, A.J. (2006). Control of ErbB signaling through metalloprotease mediated ectodomain shedding of EGF-like factors. *Growth Factors* 24, 121–136. <https://doi.org/10.1080/08977190600634373>.
94. Maruo, T., Matsuo, H., Otani, T., and Mochizuki, M. (1995). Role of epidermal growth factor (EGF) and its receptor in the development of the human placenta. *Reprod. Fertil. Dev.* 7, 1465–1470. <https://doi.org/10.1071/rd9951465>.
95. Massimiani, M., Vecchione, L., Piccirilli, D., Spitalieri, P., Amati, F., Salvi, S., Ferrazzani, S., Stuhlmann, H., and Campagnolo, L. (2015). Epidermal growth factor-like domain 7 promotes migration and invasion of human trophoblast cells through activation of MAPK, PI3K and NOTCH signaling pathways. *Mol. Hum. Reprod.* 21, 435–451. <https://doi.org/10.1093/molehr/gav006>.
96. Fang, L., Gao, Y., Wang, Z., Li, Y., Yan, Y., Wu, Z., Cheng, J.-C., and Sun, Y.-P. (2021). EGF stimulates human trophoblast cell invasion by downregulating ID3-mediated KISS1 expression. *Cell Commun. Signal.* 19, 101. <https://doi.org/10.1186/s12964-021-00783-2>.
97. Clough, B., and Frickel, E.-M. (2017). The *Toxoplasma* parasitophorous vacuole: an evolving Host-Parasite Frontier. *Trends Parasitol.* 33, 473–488. <https://doi.org/10.1016/j.pt.2017.02.007>.
98. Zhao, Y., Marple, A.H., Ferguson, D.J.P., Bzik, D.J., and Yap, G.S. (2014). Avirulent strains of *Toxoplasma gondii* infect macrophages by active invasion from the phagosome. *Proc. Natl. Acad. Sci. USA* 111, 6437–6442. <https://doi.org/10.1073/pnas.1316841111>.
99. Gutierrez, M.G. (2013). Functional role(s) of phagosomal Rab GTPases. *Small GTPases* 4, 148–158. <https://doi.org/10.4161/sgtp.25604>.
100. Prashar, A., Schnettger, L., Bernard, E.M., and Gutierrez, M.G. (2017). Rab GTPases in immunity and inflammation. *Front. Cell. Infect. Microbiol.* 7, 435. <https://doi.org/10.3389/fcimb.2017.00435>.
101. Gudgeon, J., Marín-Rubio, J.L., and Trost, M. (2022). The role of macrophage scavenger receptor 1 (MSR1) in inflammatory disorders and cancer. *Front. Immunol.* 13, 1012002. <https://doi.org/10.3389/fimmu.2022.1012002>.
102. Onyishi, C.U., Desanti, G.E., Wilkinson, A.L., Lara-Reyna, S., Frickel, E.-M., Fejer, G., Christophe, O.D., Bryant, C.E., Mukhopadhyay, S., Gordon, S., et al. (2023). Toll-like receptor 4 and macrophage scavenger receptor 1 crosstalk regulates phagocytosis of a fungal pathogen. *Nat. Commun.* 14, 4895. <https://doi.org/10.1038/s41467-023-40635-w>.
103. Peixoto, L., Chen, F., Harb, O.S., Davis, P.H., Beiting, D.P., Brownback, C.S., Ouloguem, D., and Roos, D.S. (2010). Integrative genomic approaches highlight a family of parasite-specific kinases that regulate host responses. *Cell Host Microbe* 8, 208–218. <https://doi.org/10.1016/j.chom.2010.07.004>.
104. Braun, L., Brenier-Pinchart, M.-P., Yogavel, M., Curt-Varesano, A., Curt-Bertini, R.-L., Hussain, T., Kieffer-Jaquinod, S., Coute, Y., Pelloux, H., Tardieux, I., et al. (2013). A *Toxoplasma* dense granule protein, GRA24, modulates the early immune response to infection by promoting a direct and sustained host p38 MAPK activation. *J. Exp. Med.* 210, 2071–2086. <https://doi.org/10.1084/jem.20130103>.
105. Ren, B., Schmid, M., Scheiner, M., Mollenkopf, H.-J., Lucius, R., Heitlinger, E., and Gupta, N. (2021). *Toxoplasma* and *Eimeria* co-opt the host cFos expression for intracellular development in mammalian cells. *Comput. Struct. Biotechnol. J.* 19, 719–731. <https://doi.org/10.1016/j.csbj.2020.12.045>.
106. Bottino, C., Castriconi, R., Pende, D., Rivera, P., Nanni, M., Carnemolla, B., Cantoni, C., Grassi, J., Marcenaro, S., Reymond, N., et al. (2003). Identification of PVR (CD155) and Nectin-2 (CD112) as cell surface ligands for the human DNAM-1 (CD226) activating molecule. *J. Exp. Med.* 198, 557–567. <https://doi.org/10.1084/jem.20030788>.
107. Wang, Y., Li, N., Zhang, X., and Horng, T. (2021). Mitochondrial metabolism regulates macrophage biology. *J. Biol. Chem.* 297, 100904. <https://doi.org/10.1016/j.jbc.2021.100904>.
108. Shrestha, S.P., Tomita, T., Weiss, L.M., and Orlofsky, A. (2006). Proliferation of *Toxoplasma gondii* in inflammatory macrophages in vivo is associated with diminished oxygen radical production in the host cell. *Int. J. Parasitol.* 36, 433–441. <https://doi.org/10.1016/j.ijpara.2006.01.006>.
109. Van Acker, T., Tavernier, J., and Peelman, F. (2019). The small GTPase Arf6: an overview of its mechanisms of action and of its role in Host-Pathogen interactions and innate immunity. *Int. J. Mol. Sci.* 20, 2209. <https://doi.org/10.3390/ijms20092209>.
110. de Oliveira Cardoso, M.F., Moreli, J.B., Gomes, A.O., de Freitas Zanon, C., Silva, A.E., Paulesu, L.R., Ietta, F., Mineo, J.R., Ferro, E.A., and Oliani, S.M. (2018). Annexin A1 peptide is able to induce an anti-parasitic effect in human placental explants infected by *Toxoplasma gondii*. *Microb. Pathog.* 123, 153–161. <https://doi.org/10.1016/j.micpath.2018.07.005>.
111. de Sousa, S.O., Santos, M.R.D., Teixeira, S.C., Ferro, E.A.V., and Oliani, S.M. (2022). Annexin A1: roles in placenta, cell survival, and nucleus. *Cells* 11, 2057. <https://doi.org/10.3390/cells11132057>.
112. Walther, A., Riehemann, K., and Gerke, V. (2000). A novel ligand of the formyl peptide receptor: annexin I regulates neutrophil extravasation by interacting with the FPR. *Mol. Cell* 5, 831–840. [https://doi.org/10.1016/S1097-2765\(00\)80323-8](https://doi.org/10.1016/S1097-2765(00)80323-8).
113. Lind, S., Dahlgren, C., Holmdahl, R., Olofsson, P., and Forsman, H. (2021). Functional selective FPR1 signaling in favor of an activation of

- the neutrophil superoxide generating NOX2 complex. *J. Leukoc. Biol.* 109, 1105–1120. <https://doi.org/10.1002/JLB.2HI0520-317R>.
114. Rashidi, S., Mansouri, R., Ali-Hassanzadeh, M., Muro, A., Nguewa, P., and Manzano-Román, R. (2023). The most prominent modulated annexins during parasitic infections. *Acta Trop.* 243, 106942. <https://doi.org/10.1016/j.actatropica.2023.106942>.
115. Howick, V.M., Russell, A.J.C., Andrews, T., Heaton, H., Reid, A.J., Natarajan, K., Butungi, H., Metcalf, T., Verzier, L.H., Rayner, J.C., et al. (2019). The Malaria Cell Atlas: single parasite transcriptomes across the complete *Plasmodium* life cycle. *Science* 365, eaaw2619. <https://doi.org/10.1126/science.aaw2619>.
116. Beeson, J.G., Brown, G.V., Molyneux, M.E., Mhango, C., Dzinjalama, F., and Rogerson, S.J. (1999). *Plasmodium falciparum* isolates from infected pregnant women and children are associated with distinct adhesive and antigenic properties. *J. Infect. Dis.* 180, 464–472. <https://doi.org/10.1086/314899>.
117. Reeder, J.C., Cowman, A.F., Davern, K.M., Beeson, J.G., Thompson, J.K., Rogerson, S.J., and Brown, G.V. (1999). The adhesion of *Plasmodium falciparum*-infected erythrocytes to chondroitin sulfate A is mediated by *P. falciparum* erythrocyte membrane protein 1. *Proc. Natl. Acad. Sci. USA* 96, 5198–5202. <https://doi.org/10.1073/pnas.96.9.5198>.
118. Beck, J.R., Muralidharan, V., Oksman, A., and Goldberg, D.E. (2014). PTEX component HSP101 mediates export of diverse malaria effectors into host erythrocytes. *Nature* 511, 592–595. <https://doi.org/10.1038/nature13574>.
119. Maier, A.G., Rug, M., O'Neill, M.T., Brown, M., Chakravorty, S., Szesztak, T., Chesson, J., Wu, Y., Hughes, K., Coppel, R.L., et al. (2008). Exported proteins required for virulence and rigidity of *Plasmodium falciparum*-infected human erythrocytes. *Cell* 134, 48–61. <https://doi.org/10.1016/j.cell.2008.04.051>.
120. Maier, A.G., Cooke, B.M., Cowman, A.F., and Tilley, L. (2009). Malaria parasite proteins that remodel the host erythrocyte. *Nat. Rev. Microbiol.* 7, 341–354. <https://doi.org/10.1038/nrmicro2110>.
121. Spillman, N.J., Beck, J.R., and Goldberg, D.E. (2015). Protein export into malaria parasite-infected erythrocytes: mechanisms and functional consequences. *Annu. Rev. Biochem.* 84, 813–841. <https://doi.org/10.1146/annurev-biochem-060614-034157>.
122. Davies, H., Belda, H., Broncel, M., Ye, X., Bisson, C., Introini, V., Dorin-Semblat, D., Semblat, J.-P., Tibúrcio, M., Gamain, B., et al. (2020). An exported kinase family mediates species-specific erythrocyte remodelling and virulence in human malaria. *Nat. Microbiol.* 5, 848–863. <https://doi.org/10.1038/s41564-020-0702-4>.
123. de Koning-Ward, T.F., Gilson, P.R., Boddey, J.A., Rug, M., Smith, B.J., Papenfuss, A.T., Sanders, P.R., Lundie, R.J., Maier, A.G., Cowman, A.F., et al. (2009). A newly discovered protein export machine in malaria parasites. *Nature* 459, 945–949. <https://doi.org/10.1038/nature08104>.
124. Warncke, J.D., Vakonakis, I., and Beck, H.-P. (2016). *Plasmodium* helical interspersed subtelomeric (PHIST) proteins, at the center of host cell remodeling. *Mol. Biol. Rev.* 80, 905–927. <https://doi.org/10.1128/MMBR.00014-16>.
125. Spielmann, T., Montagna, G.N., Hecht, L., and Matuschewski, K. (2012). Molecular make-up of the *Plasmodium* parasitophorous vacuolar membrane. *Int. J. Med. Microbiol.* 302, 179–186. <https://doi.org/10.1016/j.ijmm.2012.07.011>.
126. Garten, M., Nasamu, A.S., Niles, J.C., Zimmerberg, J., Goldberg, D.E., and Beck, J.R. (2018). EXP2 is a nutrient-permeable channel in the vacuolar membrane of *Plasmodium* and is essential for protein export via PTEX. *Nat. Microbiol.* 3, 1090–1098. <https://doi.org/10.1038/s41564-018-0222-7>.
127. Mesén-Ramírez, P., Bergmann, B., Tran, T.T., Garten, M., Stäcker, J., Naranjo-Prado, I., Höhn, K., Zimmerberg, J., and Spielmann, T. (2019). EXP1 is critical for nutrient uptake across the parasitophorous vacuole membrane of malaria parasites. *PLoS Biol.* 17, e3000473. <https://doi.org/10.1371/journal.pbio.3000473>.
128. Weiss, D.J., Lucas, T.C.D., Nguyen, M., Nandi, A.K., Bisanzio, D., Battle, K.E., Cameron, E., Twohig, K.A., Pfeffer, D.A., Rozier, J.A., et al. (2019). Mapping the global prevalence, incidence, and mortality of *Plasmodium falciparum*, 2000–17: a spatial and temporal modelling study. *Lancet* 394, 322–331. [https://doi.org/10.1016/S0140-6736\(19\)31097-9](https://doi.org/10.1016/S0140-6736(19)31097-9).
129. Gething, P.W., Patil, A.P., Smith, D.L., Guerra, C.A., Elyazar, I.R.F., Johnston, G.L., Tatem, A.J., and Hay, S.I. (2011). A new world malaria map: *Plasmodium falciparum* endemicity in 2010. *Malar. J.* 10, 378. <https://doi.org/10.1186/1475-2875-10-378>.
130. Goldenberg, R.L., McClure, E.M., Saleem, S., and Reddy, U.M. (2010). Infection-related stillbirths. *Lancet* 375, 1482–1490. [https://doi.org/10.1016/S0140-6736\(09\)61712-8](https://doi.org/10.1016/S0140-6736(09)61712-8).
131. Seale, A.C. (2020). Recognising the burden of maternal infection worldwide. *Lancet Glob. Health* 8, e615–e616. [https://doi.org/10.1016/S2214-109X\(20\)30126-1](https://doi.org/10.1016/S2214-109X(20)30126-1).
132. WHO Global Maternal Sepsis Study (GLOSS) Research Group (2020). Frequency and management of maternal infection in health facilities in 52 countries (GLOSS): a 1-week inception cohort study. *Lancet Glob. Health* 8, e661–e671. [https://doi.org/10.1016/S2214-109X\(20\)30109-1](https://doi.org/10.1016/S2214-109X(20)30109-1).
133. Haber, A.L., Biton, M., Rogel, N., Herbst, R.H., Shekhar, K., Smillie, C., Burgin, G., Delorey, T.M., Howitt, M.R., Katz, Y., et al. (2017). A single-cell survey of the small intestinal epithelium. *Nature* 551, 333–339. <https://doi.org/10.1038/nature24489>.
134. Mancio-Silva, L., Gural, N., Real, E., Wadsworth, M.H., 2nd, Butty, V.L., March, S., Nerurkar, N., Hughes, T.K., Roobsoong, W., Fleming, H.E., et al. (2022). A single-cell liver atlas of *Plasmodium vivax* infection. *Cell Host Microbe* 30, 1048–1060.e5. <https://doi.org/10.1016/j.chom.2022.03.034>.
135. Salahudeen, A.A., Choi, S.S., Rustagi, A., Zhu, J., van Unen, V., de la O, S.M., Flynn, R.A., Margalef-Català, M., Santos, A.J.M., Ju, J., et al. (2020). Progenitor identification and SARS-CoV-2 infection in human distal lung organoids. *Nature* 588, 670–675. <https://doi.org/10.1038/s41586-020-3014-1>.
136. Mor, G., Cardenas, I., Abrahams, V., and Guller, S. (2011). Inflammation and pregnancy: the role of the immune system at the implantation site. *Ann. N. Y. Acad. Sci.* 1221, 80–87. <https://doi.org/10.1111/j.1749-6632.2010.05938.x>.
137. Vesce, F. (2021). From pregnancy loss to COVID 19 cytokine storm: A matter of inflammation and coagulation. In *Interleukins – the Immune and Non-immune Systems' Related Cytokines* (IntechOpen).
138. Mullins, E., Hudak, M.L., Banerjee, J., Getzlaff, T., Townson, J., Barnette, K., Playle, R., Perry, A., Bourne, T., Lees, C.C., et al. (2021). Pregnancy and neonatal outcomes of COVID-19: coreporting of common outcomes from PAN-COVID and AAP-SONPM registries. *Ultrasound Obstet. Gynecol.* 57, 573–581. <https://doi.org/10.1002/uog.23619>.
139. Wei, S.Q., Bilodeau-Bertrand, M., Liu, S., and Auger, N. (2021). The impact of COVID-19 on pregnancy outcomes: a systematic review and meta-analysis. *CMAJ* 193, E540–E548. <https://doi.org/10.1503/cmaj.202604>.
140. Garcia-Flores, V., Romero, R., Xu, Y., Theis, K.R., Arenas-Hernandez, M., Miller, D., Peyvandipour, A., Bhatti, G., Galaz, J., Gershater, M., et al. (2022). Maternal-fetal immune responses in pregnant women infected with SARS-CoV-2. *Nat. Commun.* 13, 320. <https://doi.org/10.1038/s41467-021-27745-z>.
141. Argueta, L.B., Lacko, L.A., Bram, Y., Tada, T., Carrau, L., Rendeiro, A.F., Zhang, T., Uhl, S., Lubor, B.C., Chandar, V., et al. (2022). Inflammatory responses in the placenta upon SARS-CoV-2 infection late in pregnancy. *iScience* 25, 104223. <https://doi.org/10.1016/j.isci.2022.104223>.
142. Adam, I., Elhassan, E.M., Mohammed, A.A., Salih, M.M., and Elbashir, M.I. (2011). Malaria and pre-eclampsia in an area with unstable malaria transmission in Central Sudan. *Malar. J.* 10, 258. <https://doi.org/10.1186/1475-2875-10-258>.
143. Alshareef, S.A., Nasr, A.M., and Adam, I. (2018). Toxoplasma gondii infection and pre-eclampsia among Sudanese women. *Trans. R. Soc. Trop. Med. Hyg.* 112, 393–396. <https://doi.org/10.1093/trstmh/try067>.

144. Higgins, L., Vause, S., and Tower, C. (2010). Cytomegalovirus infection in association with early onset pre-eclampsia. *BMJ Case Rep.* 2010, bcr0320102803. <https://doi.org/10.1136/bcr.03.2010.2803>.
145. DePaoli Taylor, B., Hill, A.V., Perez-Patron, M.J., Haggerty, C.L., Schisterman, E.F., Naimi, A.I., Noah, A., and Comeaux, C.R. (2022). Sexually transmitted infections and risk of hypertensive disorders of pregnancy. *Sci. Rep.* 12, 13904. <https://doi.org/10.1038/s41598-022-17989-0>.
146. Mattuuzzi, A., Sauvestre, F., André, G., Poingt, M., Camberlein, C., Carles, D., Pelluard, F., Blanco, P., Sentilhes, L., and Lazaro, E. (2020). Adverse perinatal outcomes of chronic intervillitis of unknown etiology: an observational retrospective study of 122 cases. *Sci. Rep.* 10, 12611. <https://doi.org/10.1038/s41598-020-69191-9>.
147. Labarrere, C., and Mullen, E. (1987). Fibrinoid and trophoblastic necrosis with massive chronic intervillitis: an extreme variant of villitis of unknown etiology. *Am. J. Reprod. Immunol. Microbiol.* 15, 85–91. <https://doi.org/10.1111/j.1600-0897.1987.tb00162.x>.
148. Rogerson, S.J., Pollina, E., Getachew, A., Tadesse, E., Lema, V.M., and Molyneux, M.E. (2003). Placental monocyte infiltrates in response to *Plasmodium falciparum* malaria infection and their association with adverse pregnancy outcomes. *Am. J. Trop. Med. Hyg.* 68, 115–119. <https://doi.org/10.4269/ajtmh.2003.68.1.0680115>.
149. Srinivas, S.K., Ma, Y., Sammel, M.D., Chou, D., McGrath, C., Parry, S., and Elovitz, M.A. (2006). Placental inflammation and viral infection are implicated in second trimester pregnancy loss. *Am. J. Obstet. Gynecol.* 195, 797–802. <https://doi.org/10.1016/j.ajog.2006.05.049>.
150. Yao, Y., Xu, X.-H., and Jin, L. (2019). Macrophage polarization in physiological and pathological pregnancy. *Front. Immunol.* 10, 792. <https://doi.org/10.3389/fimmu.2019.00792>.
151. Tsao, F.-Y., Wu, M.-Y., Chang, Y.-L., Wu, C.-T., and Ho, H.-N. (2018). M1 macrophages decrease in the decidua from normal pregnancies but not from spontaneous abortions or unexplained recurrent spontaneous abortions. *J. Formos. Med. Assoc.* 117, 204–211. <https://doi.org/10.1016/j.fjma.2017.03.011>.
152. Radović, Janošević, D., Popović, J., Krstić, M., Tubić-Pavlović, A., Stefanović, M., and Pop-Trajković, S. (2016). The structure of immunocompetent decidual cells in recurrent missed abortions. *Vojnosanit. Pregl.* 73, 306–311. <https://doi.org/10.2298/VSP141226018R>.
153. Dubicke, A., Ekman-Ordeberg, G., Mazurek, P., Miller, L., and Yellon, S.M. (2016). Density of stromal cells and macrophages associated with collagen remodeling in the human cervix in preterm and term birth. *Reprod. Sci.* 23, 595–603. <https://doi.org/10.1177/1933719115616497>.
154. Petraglia, F., Arcuri, F., de Ziegler, D., and Chapron, C. (2012). Inflammation: a link between endometriosis and preterm birth. *Fertil. Steril.* 98, 36–40. <https://doi.org/10.1016/j.fertnstert.2012.04.051>.
155. Lima, T.S., Gov, L., and Lodoen, M.B. (2018). Evasion of human neutrophil-mediated host defense during *Toxoplasma gondii* infection. *mBio* 9, 10–1128. <https://doi.org/10.1128/mBio.02027-17>.
156. Gov, L., Schneider, C.A., Lima, T.S., Pandori, W., and Lodoen, M.B. (2017). NLRP3 and potassium efflux drive rapid IL-1 β release from primary human monocytes during *Toxoplasma gondii* infection. *J. Immunol.* 199, 2855–2864. <https://doi.org/10.4049/jimmunol.1700245>.
157. Levine, R.J., Maynard, S.E., Qian, C., Lim, K.-H., England, L.J., Yu, K.F., Schisterman, E.F., Thadhani, R., Sachs, B.P., Epstein, F.H., et al. (2004). Circulating angiogenic factors and the risk of preeclampsia. *N. Engl. J. Med.* 350, 672–683. <https://doi.org/10.1056/NEJMoa031884>.
158. Muehlenbachs, A., Mutabingwa, T.K., Edmonds, S., Fried, M., and Duffy, P.E. (2006). Hypertension and maternal-fetal conflict during placental malaria. *PLoS Med.* 3, e446. <https://doi.org/10.1371/journal.pmed.0030446>.
159. Palis, J., and Yoder, M.C. (2001). Yolk-sac hematopoiesis: the first blood cells of mouse and man. *Exp. Hematol.* 29, 927–936. [https://doi.org/10.1016/s0301-472x\(01\)00669-5](https://doi.org/10.1016/s0301-472x(01)00669-5).
160. Jagannathan-Bogdan, M., and Zon, L.I. (2013). Hematopoiesis. *Development* 140, 2463–2467. <https://doi.org/10.1242/dev.083147>.
161. Park, M.D., Silvin, A., Ginhoux, F., and Merad, M. (2022). Macrophages in health and disease. *Cell* 185, 4259–4279. <https://doi.org/10.1016/j.cell.2022.10.007>.
162. Ten Hoeve, A.L., Braun, L., Rodriguez, M.E., Olivera, G.C., Bougdour, A., Belmudes, L., Couté, Y., Saeij, J.P.J., Hakimi, M.-A., and Barragan, A. (2022). The *Toxoplasma* effector GRA28 promotes parasite dissemination by inducing dendritic cell-like migratory properties in infected macrophages. *Cell Host Microbe* 30, 1570–1588.e7. <https://doi.org/10.1016/j.chom.2022.10.001>.
163. Zhou, D.H., Yuan, Z.G., Zhao, F.R., Li, H.L., Zhou, Y., Lin, R.Q., Zou, F.C., Song, H.Q., Xu, M.J., and Zhu, X.Q. (2011). Modulation of mouse macrophage proteome induced by *Toxoplasma gondii* tachyzoites in vivo. *Parasitol. Res.* 109, 1637–1646. <https://doi.org/10.1007/s00436-011-2435-z>.
164. Simoni, M.K., Jurado, K.A., Abrahams, V.M., Fikrig, E., and Guller, S. (2017). Zika virus infection of Hofbauer cells. *Am. J. Reprod. Immunol.* 77, e12613. <https://doi.org/10.1111/aji.12613>.
165. Quicke, K.M., Bowen, J.R., Johnson, E.L., McDonald, C.E., Ma, H., O'Neal, J.T., Rajakumar, A., Wrammert, J., Rimawi, B.H., Pulendran, B., et al. (2016). Zika virus infects human placental macrophages. *Cell Host Microbe* 20, 83–90. <https://doi.org/10.1016/j.chom.2016.05.015>.
166. Muffat, J., Li, Y., Omer, A., Durbin, A., Bosch, I., Bakiasi, G., Richards, E., Meyer, A., Gehrke, L., and Jaenisch, R. (2018). Human induced pluripotent stem cell-derived glial cells and neural progenitors display divergent responses to Zika and dengue infections. *Proc. Natl. Acad. Sci. USA* 115, 7117–7122. <https://doi.org/10.1073/pnas.1719266115>.
167. Venugopal, K., Hentzschel, F., Valkiūnas, G., and Marti, M. (2020). *Plasmodium* asexual growth and sexual development in the haematopoietic niche of the host. *Nat. Rev. Microbiol.* 18, 177–189. <https://doi.org/10.1038/s41579-019-0306-2>.
168. Kho, S., Qotrunnada, L., Leonardo, L., Andries, B., Wardani, P.A.I., Fricot, A., Henry, B., Hardy, D., Margyaningsih, N.I., Apriyanti, D., et al. (2021). Hidden biomass of intact malaria parasites in the human spleen. *N. Engl. J. Med.* 384, 2067–2069. <https://doi.org/10.1056/NEJMc2023884>.
169. Del Portillo, H.A., Ferrer, M., Brugat, T., Martin-Jaular, L., Langhorne, J., and Lacerda, M.V.G. (2012). The role of the spleen in malaria. *Cell. Microbiol.* 14, 343–355. <https://doi.org/10.1111/j.1462-5822.2011.01741.x>.
170. Lee, W.-C., Russell, B., and Réna, L. (2019). Sticking for a cause: the *falciparum* malaria parasites cytoadherence paradigm. *Front. Immunol.* 10, 1444. <https://doi.org/10.3389/fimmu.2019.01444>.
171. Franke-Fayard, B., Fonager, J., Braks, A., Khan, S.M., and Janse, C.J. (2010). Sequestration and tissue accumulation of human malaria parasites: can we learn anything from rodent models of malaria? *PLoS Pathog.* 6, e1001032. <https://doi.org/10.1371/journal.ppat.1001032>.
172. Boddey, J.A., and Cowman, A.F. (2013). *Plasmodium* nesting: remaking the erythrocyte from the inside out. *Annu. Rev. Microbiol.* 67, 243–269. <https://doi.org/10.1146/annurev-micro-092412-155730>.
173. Elsworth, B., Matthews, K., Nie, C.Q., Kalanon, M., Charnaud, S.C., Sanders, P.R., Chisholm, S.A., Counihan, N.A., Shaw, P.J., Pino, P., et al. (2014). PTEX is an essential nexus for protein export in malaria parasites. *Nature* 511, 587–591. <https://doi.org/10.1038/nature13555>.
174. Francis, S.E., Malkov, V.A., Oleinikov, A.V., Rossnagle, E., Wendler, J.P., Mutabingwa, T.K., Fried, M., and Duffy, P.E. (2007). Six genes are preferentially transcribed by the circulating and sequestered forms of *Plasmodium falciparum* parasites that infect pregnant women. *Infect. Immun.* 75, 4838–4850. <https://doi.org/10.1128/IAI.00635-07>.
175. GOV (2021). Malaria imported into the UK. <https://www.gov.uk/government/publications/malaria-in-the-uk-annual-report/malaria-imported-into-the-uk-2021>.
176. GOV Listeriosis in England and Wales: summary for 2021. <https://www.gov.uk/government/publications/listeria-monocytogenes-surveillance-reports/listeriosis-in-england-and-wales-summary-for-2021>.

177. GOV (2022). Common animal-associated infections (England and Wales): fourth quarter. <https://www.gov.uk/government/publications/common-animal-associated-infections-2022/common-animal-associated-infections-england-and-wales-fourth-quarter-2022>.
178. Balestrino, D., Hamon, M.A., Dortet, L., Nahori, M.-A., Pizarro-Cerda, J., Alignani, D., Dussurget, O., Cossart, P., and Toledo-Arana, A. (2010). Single-cell techniques using chromosomally tagged fluorescent bacteria to study *Listeria monocytogenes* infection processes. *Appl. Environ. Microbiol.* 76, 3625–3636. <https://doi.org/10.1128/AEM.02612-09>.
179. Adjalley, S.H., Johnston, G.L., Li, T., Eastman, R.T., Ekland, E.H., Eappen, A.G., Richman, A., Sim, B.K.L., Lee, M.C.S., Hoffman, S.L., et al. (2011). Quantitative assessment of *Plasmodium falciparum* sexual development reveals potent transmission-blocking activity by methylene blue. *Proc. Natl. Acad. Sci. USA* 108, E1214–E1223. <https://doi.org/10.1073/pnas.1112037108>.
180. Bachmann, A., Petter, M., Krumkamp, R., Esen, M., Held, J., Scholz, J.A.M., Li, T., Sim, B.K.L., Hoffman, S.L., Kremsner, P.G., et al. (2016). Mosquito Passage Dramatically Changes var Gene Expression in Controlled Human *Plasmodium falciparum* Infections. *PLoS Pathog.* 12, e1005538. <https://doi.org/10.1371/journal.ppat.1005538>.
181. Wolf, F.A., Angerer, P., and Theis, F.J. (2018). SCANPY: large-scale single-cell gene expression data analysis. *Genome Biol.* 19, 15. <https://doi.org/10.1186/s13059-017-1382-0>.
182. Kaminow, B., Yunusov, D., and Dobin, A. (2021). STARsolo: accurate, fast and versatile mapping/quantification of single-cell and single-nucleus RNA-seq data. Preprint at bioRxiv. <https://doi.org/10.1101/2021.05.05.442755>.
183. Wolock, S.L., Lopez, R., and Klein, A.M. (2019). Scrublet: computational identification of cell doublets in single-cell transcriptomic data. *Cell Syst.* 8, 281–291.e9. <https://doi.org/10.1016/j.cels.2018.11.005>.
184. Efremova, M., Vento-Tormo, M., Teichmann, S.A., and Vento-Tormo, R. (2020). CellPhoneDB: inferring cell-cell communication from combined expression of multi-subunit ligand-receptor complexes. *Nat. Protoc.* 15, 1484–1506. <https://doi.org/10.1038/s41596-020-0292-x>.
185. Garcia-Alonso, L., Lorenzi, V., Mazzeo, C.I., Alves-Lopes, J.P., Roberts, K., Sancho-Serra, C., Engelbert, J., Marečková, M., Gruhn, W.H., Botting, R.A., et al. (2022). Single-cell roadmap of human gonadal development. *Nature* 607, 540–547. <https://doi.org/10.1038/s41586-022-04918-4>.
186. Heaton, H., Talman, A.M., Knights, A., Imaz, M., Gaffney, D.J., Durbin, R., Hemberg, M., and Lawnczak, M.K.N. (2020). Souporecell: robust clustering of single-cell RNA-seq data by genotype without reference genotypes. *Nat. Methods* 17, 615–620. <https://doi.org/10.1038/s41592-020-0820-1>.
187. Ritchie, M.E., Phipson, B., Wu, D., Hu, Y., Law, C.W., Shi, W., and Smyth, G.K. (2015). limma powers differential expression analyses for RNA-sequencing and microarray studies. *Nucleic Acids Res.* 43, e47. <https://doi.org/10.1093/nar/gkv007>.
188. Robinson, M.D., McCarthy, D.J., and Smyth, G.K. (2010). edgeR: a Bioconductor package for differential expression analysis of digital gene expression data. *Bioinformatics* 26, 139–140. <https://doi.org/10.1093/bioinformatics/btp616>.
189. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., et al. (2011). Scikit-learn: machine learning in Python. *J. Mach. Learn. Res.* 12, 2825–2830.
190. Lopez, R., Regier, J., Cole, M.B., Jordan, M.I., and Yosef, N. (2018). Deep generative modeling for single-cell transcriptomics. *Nat. Methods* 15, 1053–1058. <https://doi.org/10.1038/s41592-018-0229-2>.
191. Korsunsky, I., Fan, J., Slowikowski, K., Zhang, F., Wei, K., Baglaenko, Y., Brenner, M., Loh, P.-R., and Raychaudhuri, S. (2018). Fast, sensitive, and accurate integration of single cell data with Harmony. Preprint at bioRxiv. <https://doi.org/10.1101/461954>.
192. Radfar, A., Méndez, D., Moneriz, C., Linares, M., Marín-García, P., Puyet, A., Diez, A., and Bautista, J.M. (2009). Synchronous culture of *Plasmodium falciparum* at high parasitemia levels. *Nat. Protoc.* 4, 1899–1915. <https://doi.org/10.1038/nprot.2009.198>.
193. Nielsen, M.A., Pinto, V.V., Resende, M., Dahlbäck, M., Ditlev, S.B., Theander, T.G., and Salanti, A. (2009). Induction of adhesion-inhibitory antibodies against placental *Plasmodium falciparum* parasites by using single domains of VAR2CSA. *Infect. Immun.* 77, 2482–2487. <https://doi.org/10.1128/IAI.00159-09>.
194. Mesén-Ramírez, P., Bergmann, B., Elhabiri, M., Zhu, L., von Thien, H., Castro-Peña, C., Gilberger, T.-W., Davioud-Charvet, E., Bozdech, Z., Bachmann, A., et al. (2021). The parasitophorous vacuole nutrient channel is critical for drug access in malaria parasites and modulates the artemisinin resistance fitness cost. *Cell Host Microbe* 29, 1774–1787.e9. <https://doi.org/10.1016/j.chom.2021.11.002>.
195. Bozdech, Z., Llinás, M., Pulliam, B.L., Wong, E.D., Zhu, J., and DeRisi, J.L. (2003). The transcriptome of the intraerythrocytic developmental cycle of *Plasmodium falciparum*. *PLoS Biol.* 1, E5. <https://doi.org/10.1371/journal.pbio.0000005>.
196. Krishnaswami, S.R., Grindberg, R.V., Novotny, M., Venepally, P., Lacar, B., Bhutani, K., Linker, S.B., Pham, S., Erwin, J.A., Miller, J.A., et al. (2016). Using single nuclei for RNA-seq to capture the transcriptome of postmortem neurons. *Nat. Protoc.* 11, 499–524. <https://doi.org/10.1038/nprot.2016.015>.
197. Bayraktar, O.A., Bartels, T., Holmqvist, S., Kleshchevnikov, V., Martirosyan, A., Polioudakis, D., Ben Haim, L., Young, A.M.H., Batiuk, M.Y., Prakash, K., et al. (2020). Astrocyte layers in the mammalian cerebral cortex revealed by a single-cell in situ transcriptomic map. *Nat. Neurosci.* 23, 500–509. <https://doi.org/10.1038/s41593-020-0602-1>.
198. Stuart, T., Butler, A., Hoffman, P., Hafemeister, C., Papalexi, E., Mauck, W.M., 3rd, Hao, Y., Stoeckius, M., Smibert, P., and Satija, R. (2019). Comprehensive integration of single-cell data. *Cell* 177, 1888–1902.e21. <https://doi.org/10.1016/j.cell.2019.05.031>.
199. Zhou, Y., Zhou, B., Pache, L., Chang, M., Khodabakhshi, A.H., Tanaseichuk, O., Benner, C., and Chanda, S.K. (2019). Metascape provides a biologist-oriented resource for the analysis of systems-level datasets. *Nat. Commun.* 10, 1523. <https://doi.org/10.1038/s41467-019-09234-6>.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
CD138 (Syndecan-1) FITC Antibody, anti-human, REAfinity	Miltenyi Biotec	130-115-478; REA929; RRID: AB_2857402
EpCAM (VU1D9) Mouse mAb	Cell Signaling Technology	2929S; VU1D9; RRID: AB_2098657
Anti-Green fluorescence protein (GFP)	Abcam	ab13970; RRID: AB_300798
Anti-Listeria monocytogenes	Abcam	ab35132; RRID: AB_776048
Anti-Toxoplasma gondii antibody	Abcam	ab23507; RRID: AB_447489
CD138 (SDC1) PE Antibody	Miltenyi Biotec	130-119-840; 44F9; RRID: AB_2751879
FITC anti-human CD45	Biolegend	304006; HI30; RRID: AB_314393
LIVE/DEAD™ Fixable Near-IR Dead Cell Stain	Thermo Fisher	L10119
Mitotracker Deep Red FM	Invitrogen	M22426
Alexa Fluor® 488 anti-human CD64 Antibody	Biolegend	305010;10.1; RRID:AB_528865
Recombinant Anti-Vimentin antibody	Abcam	ab92547, EPR3776; RRID:AB_10562134
Anti-HLA G antibody [MEM-G/9] PE	Abcam	ab24384; RRID:AB_448029
Goat anti-Rabbit IgG (H+L) Cross-Adsorbed Secondary Antibody, Alexa Fluor™ 488	Invitrogen	A11008; RRID:AB_143165
Donkey anti-Goat IgG (H+L) Cross-Adsorbed Secondary Antibody, Alexa Fluor™ 647	Invitrogen	A21447; RRID:AB_2535864
Donkey anti-Rabbit IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor™ Plus 555	Invitrogen	A32794; RRID:AB_2762834
Goat Anti-Mouse IgG H&L (Alexa Fluor® 594) preadsorbed	Abcam	ab150120; RRID:AB_2631447
Bacterial and virus strains		
<i>Listeria monocytogenes</i> EGD-cGFP	Balestrino et al. ¹⁷⁸	N/A
Biological samples		
Human placental tissue	Human Developmental Biology Resource United Kingdom	http://www.hdbr.org
Critical commercial assays		
Milliplex Human Cytokine/Chemokine Magnetic Bead Panel	Merck Millipore	HCYTOMAG-60K-41
Human VEGF R1 (FLT1) ELISA Kit	Abcam	ab289705
Glucose Uptake Assay Kit-Red	Dojindo EU	UP03
Deposited data		
Placental explant single cell RNA sequencing	This paper	ArrayExpress: E-MTAB-12795
Placenta cell hashing single cell RNA sequencing	This paper	ArrayExpress: E-MTAB-12795
Infected placental explants single cell RNA sequencing	This paper	ArrayExpress: E-MTAB-12795
Infected placental explants single cell RNA sequencing, integrated and log-transformed data	This paper	https://www.reproductivecellatlas.org
Infected placental explants single nuclei RNA sequencing	This paper	ArrayExpress: E-MTAB-12795
2D9 <i>Plasmodium falciparum</i> single cell RNA sequencing	This paper	ArrayExpress: E-MTAB-12795
Maternal-fetal interface atlas	Arutyunyan et al. ⁴⁷	ArrayExpress: E-MTAB-12421
PBMC 10x datasets	https://www.10xgenomics.com/datasets/10k-human-pbmcs-3-v3-1-chromium-controller-3-1-high	10k Human PBMCs, 3' v3.1, Chromium Controller (2021, September 14)
Human reference genome GRCh38 (ver. 2020-A, CellRanger)	10x Genomics	https://cf.10xgenomics.com/supp/cell-exp/refdata-gex-GRCh38-2020-A.tar.gz

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
<i>P. falciparum</i> 3D7 genome v3	Wellcome Sanger Institute	https://www.sanger.ac.uk/resources/downloads/protozoa/
<i>T. gondii</i> GT1 genome, GCA_000149715.2	ENSEMBL	https://protists.ensembl.org/Toxoplasma_gondii_gt1_gca_000149715/
Experimental models: Cell lines		
Human foreskin fibroblast (HFF)	ATCC	ATCC-CRL-2429
Experimental models: Organisms/strains		
<i>Plasmodium falciparum</i> NF54-eGFP	Adjalley et al. ¹⁷⁹	N/A
<i>Plasmodium falciparum</i> 2D9 NF54-eGFP	This paper	N/A
<i>Plasmodium falciparum</i> 3D7-EXP1wtlow-Ty1 (sfa32) in condΔEXP1	Mesén-Ramírez et al. ¹²⁷	N/A
<i>Toxoplasma gondii</i> RH+HX (wild type)	Ross F. Waller	N/A
Oligonucleotides		
Real-time PCR primers	Bachmann et al. ¹⁸⁰	Table S5
Software and algorithms		
Python (v3.8.8)	Python	https://www.python.org
R (v4.0.4)	R project	https://www.R-project.org
Scanpy (v1.7.1)	Wolf et al. ¹⁸¹	https://scanpy.readthedocs.io/en/stable/index.html
STARsolo (v2.7.9a)	Kaminow et al. ¹⁸²	https://github.com/alexdobin/STAR/blob/master/docs/STARsolo.md
Scrublet	Wolock et al. ¹⁸³	https://github.com/swolock/scrublet
CellphoneDB (v4)	Efremova et al. ¹⁸⁴ ; Garcia-Alonso et al. ¹⁸⁵	https://github.com/ventolab/CellphoneDB
Souporcell (v2.5)	Heaton et al. ¹⁸⁶	https://github.com/wheaton5/souporcell
Limma (v3.46.0)	Ritchie et al. ¹⁸⁷	https://bioconductor.org/packages/release/bioc/html/limma.html
EdgeR (v3.32.1)	Robinson et al. ¹⁸⁸	https://bioconductor.org/packages/release/bioc/html/edgeR.html
Scikit-learn (v 0.24.2)	Pedregosa et al. ¹⁸⁹	https://scikit-learn.org/stable/
ScVI (v0.10.1)	Lopez et al. ¹⁹⁰	https://scvi-tools.org
Harmonypy (v0.0.9)	Korsunsky et al. ¹⁹¹	https://github.com/slowkow/harmonypy
Prism 10 (v10)	GraphPad Prism	https://www.graphpad.com/features
FLOWJO Software (v10.6.1)	BD Life Science	https://www.flowjo.com/solutions/flowjo
Custom code used for analysis	This paper	https://github.com/ventolab/Placenta-Infection https://doi.org/10.5281/zenodo.10881824
Other		
Homo sapiens annexin A1 (ANXA1) mRNA	ACD, Bio-Techne	465418
RNAscope 2.5 LS Probe - Hs-AREG-C3	ACD, Bio-Techne	313118-C3
RNAscope 2.5 LS Probe Hs-CCL20-C2	ACD, Bio-Techne	409618-C2
RNAscope 2.5 LS Probe Hs-CCL4-C1	ACD, Bio-Techne	455348
RNAscope 2.5 LS Probe Hs-Hs-CCR5-C4	ACD, Bio-Techne	601508-C4 (bespoke)
RNAscope 2.5 LS Probe Hs-Hs-CD14-C2	ACD, Bio-Techne	418808-C2
RNAscope 2.5 LS Probe - Hs-FPR2-C3	ACD, Bio-Techne	442008-C3
RNAscope 2.5 LS Probe Hs-GATA3-C3	ACD, Bio-Techne	403558-C3
RNAscope 2.5 LS Probe Hs-IL1B-C4	ACD, Bio-Techne	310368-C4
RNAscope 2.5 LS Probe Hs-IL1R-C1	ACD, Bio-Techne	437488

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
RNAscope® 2.5 LS Probe - Hs-LGALS3	ACD, Bio-Techne	477838
Homo sapiens MER proto-oncogene tyrosine kinase (MERTK) mRNA	ACD, Bio-Techne	530538-C2

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Roser Vento-Tormo (rv4@sanger.ac.uk).

Materials availability

This study did not generate new materials.

Data and code availability

- Single-cell and single-nuclei RNA-seq raw and aligned data have been deposited at ArrayExpress: E-MTAB-12795) and are publicly available as of the date of publication. Accession numbers are listed in the [key resources table](#). The integrated and log-transformed scRNA-seq placental infection dataset can be browsed and downloaded in <https://www.reproductivecellatlas.org>. This paper also analyzes existing, publicly available data. The accession numbers for these datasets are listed in the [key resources table](#).
- All original code has been deposited at Github (<https://github.com/ventolab>) and Zenodo (<https://doi.org/10.5281/zenodo.10881824>) and is publicly available as of the date of publication. DOIs are listed in the [key resources table](#).
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Samples

All tissue samples were obtained with written informed consent from all participants in accordance with the guidelines in the Declaration of Helsinki 2000. Placental samples from elective terminations were provided by the Joint MRC/Wellcome Trust (grant# MR/R006237/1) Human Developmental Biology Resource (HDBR, <http://www.hdbbr.org>), with appropriate maternal written consent and approval from the Fulham Research Ethics Committee (REC reference 18/LO/0822). The HDBR is regulated by the UK Human Tissue Authority (HTA: www.hta.gov.uk) and operates in accordance with the relevant HTA Codes of Practice. Any subsequent secondary publications arising from our data generated from HDBR material should similarly acknowledge the HDBR.

Placental explant culture

Samples of placental tissue were collected from surgical terminations between 5 and 14 post-conceptual weeks (PCW). The developmental stage of each sample is specified in [Table S1](#). Donor ethnicity is not shared by HDBR to protect donor confidentiality. The sex of the placental samples is unknown. After collection, samples were kept on ice in Ham's/F12 media, shipped to the institute and processed within eight hours of collection. Samples were washed in ice-cold Ham's/F12. The whole sample was cleaned of blood clots and the villi identified. The area containing the most villi was cut using a scalpel to a size of approximately 1 cm². ThinCert™ 0.4 μm pore diameter 6-well plate inserts (Greiner bio-one, 657640) were pre-coated with 50% Matrigel® (Corning, CLS356255) in ice-cold Trophoblast Organoid Media (TOM)³⁶ supplemented with 10% FBS (see below for media composition). After Matrigel polymerisation, placental explants were placed on 6-wells ThinCerts™ (Greiner Bio), and the bottom well plate was filled with warm TOM (Advanced DMEM/F12 (Gibco, 12634-010), 1X B27-Vitamin A (Life Technologies, 12587010), 1x N2 (Life Technologies, 17502048), 100 μg/mL Primocin (Invivogen, ant-pm-1), 1.25 mM N-Acetyl-L-cysteine (Sigma-Aldrich, A9165-5G), L-glutamine (Sigma-Aldrich, 25030-024), 50 ng/mL recombinant human EFG (Peprotech, AF-100-15), 1.5 μM CHIR99021 (Tocris, 4423), 80 ng/mL recombinant human R-spondin-1 (R&D Systems, 4645-RS-01M/CF), 100 ng/mL recombinant human FGF-2 (Peprotech, 100-18B), 50 ng/mL recombinant human HGF (Peprotech, 100-39), 500nM A83-01 (Tocris, 2939), 2.5 μM prostaglandin E₂ (Sigma-Aldrich, P0409), 2 μM Y-27632 (Millipore, 688000) and placed in the incubator at 37°C (21% O₂, 5% CO₂). Explants were infected after ~12 h (hour) in culture and processed after 24 and 48 h post infection. For intra sample variability detection, explants were processed after 24 h in culture.

Culture of *L. monocytogenes*

A single colony of *L. monocytogenes* EGD-cGFP strain (EGD strain with chromosomally integrated GFP)¹⁷⁸ was picked from streaked Brain Heart Infusion (BHI) agar plate and inoculated in 10 ml of BHI broth. BHI broth was cultured overnight for 15 h in slanting position

at 37°C at 120 rpm. Broth optical density at 600nm (OD_{600nm}) was measured using a spectrophotometer (Biowave CO8000) and rediluted to OD_{600nm} 0.1. *L. monocytogenes* was then cultured in fresh BHI for another 2–3 h until mid-exponential growth phase prior to placental explant infection.

Culture of *T. gondii* and tachyzoite isolation

Human foreskin fibroblasts (HFFs) (ATCC, ATCC-CRL-2429) were maintained at 37°C (21% O₂, 5% CO₂) in D10 media (DMEM, high glucose (Gibco, 41965062), 10% FBS, 1% L-glutamine (ThermoFisher Scientific, 10500064), 0.5% penicillin/streptomycin and 0.1% Amphotericin B (Thermo, 15290026)) and passaged by using Trypsin-EDTA 0.25% (Gibco, 25200072). Confluent HFFs were infected with the RH+HX strain of *T. gondii* in ED1 media at 37°C (DMEM, high glucose, 1% FBS, 0.1% L-glutamine, 0.5% Pen/Strep, and 0.1% Amphotericin B). The *T. gondii* culture was propagated 2–3 times a week by infecting a new batch of HFFs.

To isolate and synchronise tachyzoites, the media of infected HFF cultures of ~48 hpi (with visible parasitic vacuoles) was removed and cells were washed 2 times with 1x PBS to remove extracellular parasites. D10 media was added to the HFFs, a cell scraper was used to detach the cells. The detached cells were transferred into a syringe with a 29 gauge needle and poured into a 15 mL falcon tube. The pressure created while passing the cells through the needle bursts the cells and releases the intracellular parasites. The parasite/host suspension was filtered using a 20 µm filter and then centrifuged at 100g for 5 min (minutes) to pellet the host cell debris. The supernatant was centrifuged at 900g for 15 min to pellet the isolated tachyzoites.

Culture of human malaria parasite *Plasmodium falciparum* and *in vitro* selection procedures

P. falciparum NF54-eGFP,¹⁷⁹ *P. falciparum* 2D9 NF54-eGFP and *P. falciparum* 3D7-EXP1wt^{low}-Ty1 (*sfa32*)¹²⁷ in condΔEXP1 were maintained in 3% hematocrit in human O+ blood in RPMI 1640 supplemented with 5% Albumax-II and 50 µg/ml hypoxanthine (Gibco), 25mM HEPES (Sigma) and 1x L-Glutamax (Gibco) in a specific mix gas containing 1% O₂, 3% CO₂ and 96% N₂. Human O+ blood was provided by the NHS Blood and Transplant, Cambridge, UK and was used with approval from the NHS Cambridge-shire Research Ethics Committee and the Wellcome Sanger Institute Human Materials and Data Management Committee. NF54-eGFP were synchronised at high parasitemia levels by 70% Percoll (17-0891-01, VWR) gradient and 5% sorbitol lysis as reported previously.¹⁹²

In vitro selection for VAR2CSA^{high} was done by repeated panning of trophozoite-infected erythrocytes (IE) on 150mm petri-dishes pre-coated with 40 µg/ml of chondroitin sulfate A (CSA) sodium salt from bovine trachea (Merck).¹⁹³ The resulting parasite line after repeated panning was designated as 2D9 PfNF54-eGFP. We used the 2D9 PfNF54-eGFP for all subsequent placental explant infection assays.

Ring stage parasites of 3D7-EXP1wt^{low}-Ty1 (*sfa32*) in condΔEXP1 were synchronised twice with 5% sorbitol lysis within a 5 h developmental stage window. Parasite cultures were split into two flasks and 250 nM rapalog (AP21967, Clontech) was added to one of the two flasks to induce excision of endogenous EXP1, resulting in EXP1^{low} parasites with an episomal copy of EXP1 driven by the *sfa32* promoter.^{127,194} The flask without rapalog treatment served as a wild-type control culture. The medium containing rapalog was changed daily and the parasites further cultured in the presence of rapalog for 18–24 h where trophozoite stages were used for the glucose uptake assay.

METHOD DETAILS

var2csa expression analysis by quantitative Real-Time PCR

Total RNA from synchronised ring stage 2D9 PfNF54-eGFP and unpanned bulk PfNF54-eGFP cultures were isolated using Trizol-chloroform (Invitrogen) as previously reported.¹⁹⁵ The total RNA was further purified using on-column PureLink® DNase Treatment (12185010, Invitrogen) according to the manufacturer's instructions. 1 µg of purified RNA was subjected to cDNA synthesis with SuperScript III First-Strand synthesis system (18080051, Invitrogen) and primed with Oligo(dT)20 primers (18418020, Invitrogen), random hexamers (48190011, Invitrogen) and dNTP mix (18427013, Invitrogen) in a 20 µl reaction volume according to the manufacturer's recommendation. A cDNA synthesis reaction without reverse transcriptase enzyme was included as control. 10-fold diluted cDNA was used for quantitative Real-time PCR with Power SYBR Green PCR Master Mix (4367659, Invitrogen) master mix with 500nM forward and reverse optimised primer pairs¹⁸⁰ (Table S5) in a final volume of 20 µl. Reactions were run on the StepOnePlus Real-Time PCR Systems (Thermo Fisher Scientific) at default parameters; 95°C for 10 min, then subjected to 40 cycles of 95°C for 15 s and 60°C for 1 min, with a subsequent melting step. Transcriptional level was normalized against the housekeeping gene *PF3D7_1218600* (arginine-tRNA ligase). Relative gene expression was calculated by 2^{−ΔΔCt} method, with a different 2D9 PfNF54-eGFP panned cycles compared to unpanned bulk PfNF54-eGFP culture. Expression of *var2csa* was induced up to 50-fold after pan cycle 4, and maintained its expression after pan cycle 12, whereas expression of other VAR genes remained stable throughout the panning cycles (Figure S3B). This result supports the expansion of a clonal VAR2CSA^{high} specific population, which we termed as 2D9 PfNF54-eGFP. In addition, we also confirmed that the 2D9 VAR2CSA^{high} parasite population maintains endogenous expression of GFP at trophozoite stage (Figure S3C). We then utilised the synchronised ring stage 2D9 VAR2CSA^{high} parasites in all of our subsequent malaria infection assays.

Infection of human placental tissue explants

Infections were performed 12 h after the start of the *ex vivo* placental explant culture. All explants were washed three times with warm 1xPBS prior to infection with pathogens. Infection was done in duplicate wells per donor tissue, where one well was processed for single-cell isolation and the other was cryopreserved for single-nuclei isolation. After infection, culture supernatant was collected and snap frozen for each of the pathogen infection and time-points.

L. monocytogenes

Mid-exponential phase *L. monocytogenes* culture was washed twice with TOM/10%FBS without primocin. Placental explants were infected with 1.5×10^8 CFU (colony forming unit) of *L. monocytogenes*, an inoculum based on published literature⁴³ of infection of placental villus cultures, for 5 h in TOM, without primocin antibiotic. The explants were then washed three times with 1x PBS before adding in TOM with 50 μ g/ml of gentamicin (15750060, Thermo Fisher) for 24 and 48 h.

P. falciparum

Pf 2D9NF54-eGFP early ring-stage parasitemia was enumerated by methanol-fixed blood smear and Giemsa staining prior to adjustment for placental explant infection. Placental explants were infected with 300,000 parasites per μ L of erythrocytes, selected according to the World Health Organization (WHO) definition of hyperparasitemia ($> 250,000$ parasites per μ L of erythrocytes) in areas of high malaria transmission.⁴⁹ 20–30 μ L of early ring-stage parasites with $>4\%$ parasitemia were washed twice with TOM and pelleted by centrifugation at 1000g for 5 min, then resuspended in TOM with 5% KnockOut Serum Replacement (KSR) containing lipid-rich albumin (AlbuMAX) (Gibco, 10828028) at a final 3% hematocrit. The explants were washed three times with 1x PBS before adding in TOM with 5% ring-stage parasites. Placental explants with or without parasites were maintained at 37°C with mammalian culture gas mix 21% O₂, 5% CO₂ for 24 and 48 h. In parallel, 'pf_iv' controls were maintained in a 6-well tissue culture plate in TOM with 5% KSR with mammalian culture gas mix 21% O₂, 5% CO₂ for 24 and 48 h.

T. gondii

All infections with *T. gondii* were performed using TOM/10%FBS without A83-01 (TOM–A83), as we found that this compound impedes the infection process by restricting the parasite replication (data not shown). Placental explants were infected with 1×10^6 isolated tachyzoites according to published literature⁵⁰ for 5 h in TOM–A83. Then, the explants were washed three times with 1x PBS, TOM–A83 was added to the washed explants for 24 and 48 h.

Processing of placental tissue explants for scRNA-seq

Placental tissue explants were washed three times with 1x PBS after infection time points. Explants were treated with Corning® Cell recovery solution (Corning, CLS354253) for 15 min on ice to remove Matrigel® remnants. The plate was shaken every 5 min to dislodge the explant from the membrane. The tissue was transferred to a 50 mL Falcon tube with ice-cold PBS and washed by inverting, then transferred to a Petri dish with fresh ice-cold PBS, and the villi were gently scraped using forceps and scalpel. Using a Pasteur pipette with a cut tip, all of the tissue was transferred to a Falcon tube with 40 mL of pre-warmed 0.2% trypsin-EDTA and sealed with parafilm. Digestion was performed at 37°C on a rotator with gentle rotation at 30 rpm for 7 min and stopped with 2 mL of PBS. The tissue was filtered using a 100 μ m strainer and the strainer was washed with 5–10 mL of RT PBS (the tissue on the filter was kept for the second step of digestion). The filtrate was centrifuged at 200g for 5 min at RT. The supernatant was discarded and the pellet kept on ice (digest 1). The undigested tissue was retrieved from the filter by forceps and placed in a 50 mL Falcon tube containing 12.5 mL of prewarmed 1mg/ml Collagenase V (Sigma-Aldrich, C9263) and 40 μ g/mL DNase I (Roche, 10104159001) in PBS. The tube was sealed with Parafilm and transferred to a tube rotator in the incubator. The tissue was digested for 30 min with slow rotation at 30 rpm. The cell suspension was filtered through a 100 μ m filter and the filter was washed with 5 mL of RT PBS. The filter with any undigested tissue was discarded. Cells were pelleted at 200g for 5 min at RT (digest 2). Cells from digest 1 and digest 2 were pooled and washed with PBS with 20 μ g/mL of DNase I (DNase I buffer). Cell suspension was centrifuged at 200g for 5 min at RT, the pellet resuspended in 1 mL of DNase I buffer and kept on ice. Cells were counted using a haemocytometer.

To remove erythrocytes and debris, a Pancoll gradient was used. The cell suspension was topped up to 10 mL with DNase I and very slowly added on top of 5 mL of Human Pancoll (PAN-Biotech, P04-60500) in a 15 mL Falcon tube. The suspension was centrifuged at 600g for 20 min at RT, with using acceleration=0, break=1 settings on the centrifuge. Cells were collected from the phase interface using a Pasteur pipette. Cells were washed with DNase I buffer and centrifuged at 200g for 5 min at RT twice. Cells were passed through a 35–40 μ m filter and counted (bulk cells). 1 million of bulk cells were collected for flow cytometry assessment (see Flow cytometry section).

To enrich the final cell suspension for the CD45⁺ population, 1 million of bulk cells were separated and kept on ice and the rest was submitted to the MACS-based enrichment following the CD45 MicroBeads protocol (130-045-801, Miltenyi Biotec). The cells were centrifuged at 200g for 5 min at RT and the pellet resuspended in DNase I buffer and counted (CD45 cells). Bulk cells and CD45 cells were mixed in 1:1 ratio in PBS/0.04% BSA prior to loading of the 10x Genomics Chromium inlet. For *L. monocytogenes* infection, viable intracellular bacteria counts were enumerated by CFU from hypotonic lysis of remaining cells, which were determined by serial dilution plating on BHI agar.

Three independent placental tissue regions per donor were excised and processed to single cell suspension to perform cell hashing. For hashtag antibody staining, 500,000 cells from bulk and CD45⁺ enriched fraction were individually re-suspended in Cell Staining Buffer (420201, Biolegend) and blocked with 5 μ L Human TruStain FcX™ (422301, Biolegend) for 10 min at 4°C. Each of the cell suspensions originating from different placental tissue regions were hashtagged with different TotalSeq-B™ antibodies for 30 min at 4°C. 1 μ g of the following TotalSeq-B™ antibodies were used; TotalSeq™-B0253 anti-human Hashtag 3 Antibody (394635,

Biolegend), TotalSeq™-B0254 anti-human Hashtag 4 Antibody (394637, Biolegend) and TotalSeq™-B0255 anti-human Hashtag 5 Antibody (394639, Biolegend). Antibody-tagged cells were washed three times with 3 mL of Cell Staining Buffer using centrifugation speed of 500g for 5 min at 4°C. Bulk cells and CD45 cells were mixed in 1:1 ratio then filtered through a 40 µm cell strainer. Cells were counted and diluted to desired loading concentration to recover between ~4000 cells. Hashtagged cells were pooled at equal cell number ratio (targeted for a total of ~12000 cells) right before loading of the 10x Genomics Chromium inlet.

Isolation of *P. falciparum* 2D9-IEs from the placental tissue explants for scRNA-seq

Placental tissue explants infected with 2D9 PfNF54eGFP were washed on a rotating platform five times with 1xPBS to remove unbound parasites, which was designated as 'pf_nb'. Washed placental tissue explants were then cut into smaller pieces, followed by another gentle washing with 50mM EDTA (15575020, Invitrogen) on a rotating platform for 1 h, as previously described.¹¹⁶ The EDTA washed fraction, containing potentially cytoadherent population, was filtered through a 40 µm cell strainer to remove placental tissue pieces. The filtered fraction containing mostly bound parasites, designated as 'pf_b', was centrifuged at 1500 rpm and washed five times with 1xPBS. 2D9 PfNF54eGFP used for infection assay was also cultured in parallel in the same batch of TOM media, but without the exposure to placental tissue explants. This batch of culture was designated as 'pf_iv' which was also harvested and washed five times with 1xPBS. All three parasite fractions, 'pf_b', 'pf_nb' and 'pf_iv' were then resuspended in PBS/0.04% BSA at 3% hematocrit for flow cytometry assessment and loading of the 10x Genomics Chromium inlet.

Infection of CSA-coated petri dishes with *P. falciparum* 2D9-IEs

Synchronised 2D9 PfNF54eGFP ring-stage parasites at >4% parasitemia were resuspended in TOM with 5% KSR with mammalian culture gas mix 21% O₂, 5% CO₂. The ring-stage parasites were added on a CSA-coated dish to mimic placental explant infection for 24 h. After infection, 2D9 PfNF54eGFP was washed five times with 1xPBS to remove unbound parasites, which was designated as 'pf_nb'. A final gentle washing with 50mM EDTA (15575020, Invitrogen) was performed to release the CSA-bound fraction designated as 'pf_b'. The parasites used for this assay were cultured in parallel in the same batch of TOM media in another petri dish, without CSA, designated as 'pf_iv'. All parasite fractions 'pf_b', 'pf_nb' and 'pf_iv' were collected in a tube and washed twice with 1xPBS prior to glucose uptake assay and flow cytometry.

Processing of samples for snRNA-seq

Single-nuclei suspensions were isolated either from fresh frozen placental explants or OCT frozen tissue sections (8 x 50 µm cryosections) using a previously described protocol.¹⁹⁶ Isolated nuclei were stained with Trypan blue and counted with a haemocytometer. Nuclei suspension was then resuspended in PBS/1% BSA diluted to targeted final concentration to recover between ~5,000-6000 nuclei per sample prior to loading of the 10x Genomics Chromium inlet.

10x Genomics Chromium gene expression library preparation and sequencing

Cells and nuclei were loaded into the 10x Genomics Chromium inlet according to the manufacturer's instructions. The Chromium Next GEM Single Cell 3' v3.1 (Dual index) kit (10x Genomics) was used and libraries were prepared according to manufacturer's instructions aiming to recover between 2,000 and 25,000 cells per reaction. For cell hashing workflow, Chromium Next GEM Single Cell 3' v3.1 with Feature Barcoding technology for Cell Surface Protein was used (3' Feature Barcode kit) according to the manufacturer's protocol (10x Genomics). Sequencing was done at a minimum coverage of 20,000 read pairs per cell for gene expression libraries and 5000 read pairs per cell for cell surface protein libraries on an Illumina Novaseq S4 platform.

Flow cytometry

Placental tissue

Bulk single cell suspension was filtered through a 40 µm cell strainer, then washed in the fluorescence-activated cell sorting (FACS) buffer (2%FBS/2mM EDTA in PBS). Cells were blocked with Human TruStain FcX™ (422301, Biolegend) for 10 min at 4°C. Cells were stained with 1:40 FITC anti-human CD45 antibody clone HI30 (304006, Biolegend) and 1:400 LIVE/DEAD™ Fixable Near-IR Dead Cell Stain (L10119, Thermo Fisher) for 20 min at 4°C. Cells were washed twice with FACS buffer by centrifugation at 5 min 200g at 4°C before a final resuspension in 300µL of FACS buffer. Flow cytometry was performed using the CytoFLEX cytometer (Beckman Coulter). Flow cytometry datasets were analysed by FLOWJO Software v10.6.1 (BD Life Science).

Malaria parasites

5µL of IE at 3% hematocrit with endogenous GFP expression were labelled with 95µL staining solution containing 200nM of Mitotracker Deep Red FM (M22426, Invitrogen) and/or 100nM of SYBR Green (S7563, Thermo Fisher) for 20 min at 37°C. Mitotracker dye stock was dissolved in DMSO at a concentration of 1mM prior dilution to working concentration. Uninfected erythrocytes were included as background gating control. Flow cytometry was performed using the CytoFLEXcytometer (Beckman Coulter). Flow cytometry datasets were analysed by FLOWJO Software v10.6.1 (BD Life Science).

Tissue cryopreservation

Placental tissues and explants from control and infected with *P. falciparum*, *L. monocytogenes* and *T. gondii* were embedded in Tissue-Tek® Optimum Cutting Temperature (OCT) medium (4583, VWR) and flash frozen using a dry ice-isopentane slurry. Samples were stored at -80°C until further processing.

Immunostaining and visualisation

Whole-mount placental explants from control and infected with *P. falciparum*, *L. monocytogenes* and *T. gondii* were fixed in 4% paraformaldehyde in PBS for 45 min at RT. Placental tissue sections from OCT embedding were fixed with 4% paraformaldehyde for 30 min at RT. Placental explants, tissue sections were permeabilized and blocked with 2% Triton-X-100 with 10% FBS at RT for 1 h. Whole-mount and tissue sections were stained with primary antibodies (Table S6) in 0.02% Triton-X-100 in 2% FBS overnight at 4 °C. Matched secondary antibodies (Table S6) species (if using unconjugated primary) were diluted (Table S6) in 0.02% Triton-X-100 and incubated at RT for 3 h. Whole mount and sections were washed three times with 1x PBS. DAPI (4',6-Diamidino-2-Phenylindole, Dilactate) nuclear dye (564907, BD) was diluted to a concentration of 2 µg/ml in PBS and added for 30 min before imaging. Whole mount and sections were washed three times with 1x PBS. Whole mounts were mounted in the µ-Dish 35 mm chamber system (81156, Ibidi). Tissue sections were air-dried and mounted using ProLong™ Gold Antifade Mountant (P36930, Invitrogen). Imaging was performed using the Multiphoton DLS SP8 confocal microscope (Leica Microsystem) and process using LAS-X Application Suite X Software (Leica Microsystem) and ImageJ v1.52.

Multiplexed smFISH and high-resolution imaging

Tissue section staining and fluorescent imaging were conducted largely as described previously.¹⁹⁷ Sections were cut from fixed frozen samples embedded in OCT at a thickness of 10 or 11 µm using a cryostat, placed onto SuperFrost Plus slides (VWR) and stored at –80 °C until stained. Tissue sections were then processed using a Leica BOND RX to automate staining with the RNAscope Multiplex Fluorescent Reagent Kit v2 Assay (Advanced Cell Diagnostics, Bio-Techne), according to the manufacturers' instructions. Probes are listed in Table S7. Before staining, human fresh frozen sections were post-fixed in 4% paraformaldehyde in PBS for 15 min at 4 °C, then dehydrated through a series of 50, 70, 100 and 100% ethanol, for 5 min each. Following manual pretreatment, automated processing included epitope retrieval by protease digestion with Protease III for 10 min before probe hybridization. Tyramide signal amplification with Opal 520, Opal 570 and Opal 650 (Akoya Biosciences) and TSA-biotin (TSA Plus Biotin Kit, Perkin Elmer) and streptavidin-conjugated Atto 425 (Sigma Aldrich) was used to develop RNAscope probe channels.

Stained sections were imaged with a Perkin Elmer Opera Phenix High-Content Screening System, in confocal mode with 1 µm z-step size, using a 20× (numerical aperture (NA) 0.16, 0.299 µm per pixel) or objective. Channels were as follows: DAPI (excitation 375 nm, emission 435–480 nm), Atto 425 (excitation 425 nm, emission 463–501 nm), Opal 520 (excitation 488 nm, emission 500–550 nm), Opal 570 (excitation 561 nm, emission 570–630 nm) and Opal 650 (excitation 640 nm, emission 650–760 nm).

Protein immunoassays

Luminex immunoassays were performed using the Milliplex Human Cytokine/Chemokine Magnetic Bead Panel (HCYTOMAG-60K-41, Merck Millipore) according to the manufacturer's instructions. Cell culture supernatants were run in duplicate wells and read on the FLEXMAP 3D instrument (Thermo Fisher). Readouts were analysed using Belysa® analysis software (Thermo Fisher) to evaluate cytokine/chemokine concentrations according to the spline curve-fitting method.

To detect targeted secreted proteins in the explant supernatant, we used VEGFR1 (FLT1) (Abcam, ab289705) ELISA kits. We diluted the samples and performed the assays in duplicates as per manufacturer's instructions. Absorbance was measured using the Multiskan SkyHigh Microplate spectrophotometer (ThermoFisher Scientific). Results were analysed by Prism 10. Statistical significance between the control and infected samples was assessed with Wilcoxon matched-pairs signed rank test (* p < 0.05).

Glucose uptake assay

Trophozoite stage parasites 'pf_b', 'pf_nb' and 'pf_iv' isolated from the placental explant infection assay and CSA-coated petri dish were washed twice with glucose-free RPMI 1640 media (11879020, Thermo Fisher). Parasites were adjusted to 3% final hematocrit in glucose-free media containing 1:10000 dilution of glucose probe labelled with red fluorescence (UP03-10, Dojindo EU) for 15 min at 37°C protected from light. Unlabelled parasites were included as a control. The parasites were then washed twice with an ice-cold WI buffer (UP03-10, Dojindo EU) and then co-stained with SYBR Green for 15 min at 37°C protected from light. Flow cytometry was performed using the CytoFLEXcytometer (Beckman Coulter). Flow cytometry datasets were analysed by FLOWJO Software v10.6.1 (BD Life Science).

QUANTIFICATION AND STATISTICAL ANALYSIS

Alignment and quantification of explant sc or snRNA-seq data, doublet detection and clustering

The sequencing data from 10x Genomics was aligned and quantified using STARsolo v2.7.9a¹⁸² against the GRCh38 (ver. 2020-A, Cell Ranger) genome. For *P. falciparum* infection assay, sequencing data was aligned to a combined reference of human GRCh38 and *P. falciparum* 3D7 genome v3 (<https://www.sanger.ac.uk/resources/downloads/protozoa/>). For *T. gondii* infection assay, a combined reference of human GRCh38 and *T. gondii* GT1 genome GCA_000149715.2 (Ensembl). Sample and alignment metadata can be found in Tables S1 and S2. Gene expression (GEX) doublet scoring was done with Scrublet,¹⁸³ SoupCell¹⁸⁶ was used to flag doublets by genotype for multiplexed samples. For *T. gondii* infected explants, *T. gondii* counts were quantified per cell. Multiplexed

samples were deconvoluted using Souporecell with a $k=2$. Cells with >450 genes expressed, >1000 unique molecular identifier (UMI) counts and with mitochondrial genes <20% were kept. Nuclei with >1000 genes expressed, >2000 UMI counts and with mitochondrial genes <5% were kept. Souporecell ($k=2$, and $k=4$) was used to deconvolve the maternal and fetal origin of nuclei/cells. To estimate the cell cycle phase of each cell (G1, S or G2/M), we used the expression of G2/M and S phase markers and classified the barcodes using the method implemented in Scanpy v1.7.1¹⁸¹ `score_genes_cell_cycle` function. Principal component analysis (PCA) was used for dimensionality reduction, neighbour identification and Leiden clustering were run. Cells not expressing human genes, and with high expression of *Toxoplasma* markers were flagged as Tg tachyzoites and excluded from the downstream analysis. The samples from 24hpi and 48hpi were integrated separately using scVI v0.10.1¹⁹⁰ (batch= donor+sample, raw counts) using 30 latent variables for single-cell data and 20 latent variables for single-nuclei. Leiden clustering and uniform manifold approximation and projection (UMAP) were used for visualisation.

ScRNA-seq and snRNA-seq annotation and analysis of placental explants data

Cells were annotated using the predictions of a logistic regression model built with the 3000 top highly variable genes identified using `highly_variable_genes` Seurat v3¹⁹⁸ from Scanpy v1.7.1, and our prior knowledge of marker genes. Cells from 24hpi and 48hpi were annotated separately. The model was built using as training set the maternal-fetal interface scRNA-seq and snRNA-seq integrated dataset of 18 donors, previously generated by our group.⁴⁷ The model used to classify scRNA-seq data from explants was trained using the complete dataset with all the cell compartments ($n= 325,665$ cells; trophoblast, myeloid, endothelial, fibroblast, epithelial and lymphocyte compartments. ArrayExpress: E-MTAB-12421). For snRNA-seq data from explants, we only used the trophoblast compartment as training set ($n= 75,023$ cells; SCT, VCT_fusing, VCT, VCT_p, VCT_CCC, EVT_1, EVT_2, iEVT and eEVT). The models were built using scaled log-normalized counts for training and test sets, `max_iter = 5000`. Clusters with a high doublet score or high proportion of donor doublets tagged by Souporecell were excluded. Low QC clusters with low gene counts, high expression of apoptosis markers, and no cluster-specific genes, were removed as well. Zoom-in steps in all major cell compartments were performed, where the steps mentioned above were repeated in each compartment to increase granularity. Differential gene expression analysis between infected vs uninfected samples was done with limma (limma v3.46.0¹⁸⁷ and edgeR v3.32.1¹⁸⁸) with a matrix design addressing donor-specific effects correction. Genes with a FDR (Benjamini-Hochberg) <0.05, were considered for downstream analysis. Gene Ontology (GO) enrichment analysis was done using the differentially expressed genes as input for Metascape.¹⁹⁹ Cells from *T. gondii*-infected samples with a higher proportion of parasite counts than the expected by chance in uninfected samples were annotated as having intracellular parasites, "*Tg-intracellular*". Those cells from infected samples which did not fulfil the threshold were tagged as "*No-intracellular Tg*". Single-cell datasets from 24hpi and 48hpi were integrated into a single manifold using scVI v0.10.1¹⁹⁰ (batch= donor+sample, raw counts) with 30 latent variables.

ScRNA-seq annotation and analysis of peripheral blood mononuclear cells (PBMCs)

The PBMCs dataset *10k Human PBMCs*, 3' v3.1, *Chromium Controller* (2021, September 14), was downloaded from the 10x Genomics website (<https://www.10xgenomics.com/datasets/10k-human-pbmc-3-v3-1-chromium-controller-3-1-high>). The cells used to make this data were from a healthy female donor aged 25-30. The single cell gene expression dataset was made using the 3' v3.1, *Chromium Controller* and sequenced on an Illumina NovaSeq 6000 to a read depth of approximately 30,000 mean reads per cell. Data was processed by Cell Ranger 6.1.0, 10x Genomics. Data was analysed using Scanpy v1.7.1.¹⁸¹ Cells with >200 genes expressed and with mitochondrial genes <10% were kept. Genes expressed in less than 3 cells were removed. Principal component analysis (PCA) was used for dimensionality reduction and neighbour identification using $n_pcs=30$. Leiden clustering and uniform manifold approximation and projection (UMAP) were used for visualisation. Annotation was performed using marker genes already reported in the literature.

Cell-cell communication predictions

Ligand-receptor interactions were inferred using the repository of curated ligand-receptor interactions CellPhoneDB v4 (Tables S3 and S4).^{184,185} Log-normalized counts and the DEGs between infected vs uninfected samples were used to run the method `deg_analysis` from CellPhoneDB v4 with the parameters `-counts-data hgnc_symbol, -threshold 0.1`. Relevant interactions were annotated using CellPhoneDB and by manual curation. The curation process addressed for soluble and membrane molecules.

Placenta hashing scRNA-seq analysis

For hashing samples, both libraries from hashing and GEX samples were processed and aligned together with Cellranger v3.1. Antibody capture/HTO reads with fewer than 500 unique molecular identifier counts (UMI) and doubles were excluded. Cell barcodes and hash barcodes were merged to create a GEX matrix with only overlapping barcodes, with each cell containing the HTO identifier. The GEX matrix was then processed according to Figure S2A.

P. falciparum scRNA-seq analysis

For malaria parasite reads, low-quality cells with <200 genes and <300 UMI counts were removed. Reads mapping to humans and parasites were identified. Total read counts per cell mapping to >90% of the *P. falciparum* genome were included in the analysis. The samples were integrated using *harmonypy* v0.0.9¹⁹¹ (batch=parasite batch). Leiden clustering and UMAP were used for

visualisation. Cell clusters were annotated based on prior knowledge of intra-erythrocytic life cycle marker genes and Term Frequency-inverse Document Frequency (TF-IDF, scikit-learn package¹⁸⁵) ordering for top marker genes for each Leiden cluster. Differential gene expression analysis between 'pf_b' and 'pf_iv' samples was done with limma (limma v3.46.0¹⁸⁷ and edgeR v3.32.1¹⁸⁸) with a matrix design addressing batch-specific effects correction. Genes with a FDR (Benjamini-Hochberg) <0.05, were statistically significant.

ADDITIONAL RESOURCES

The scRNA-seq dataset of infected placenta to reproduce the UMAP and dotplots can be accessed, visualised and downloaded through the website portal <https://www.reproductivecellatlas.org>.